Finite-Dimensional Type I von Neumann Algebras in PyTorch: A GPU-Accelerated Framework for Random Block-Diagonal Operators

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Abstract

Finite-dimensional von Neumann algebras, the direct sums $\bigoplus_c M_{k_c}(\mathbb{C})$ of matrix algebras, are basic objects of operator algebra theory and, through the commutant of a symmetry, the natural home of quantum systems with conserved charges or superselection rules. To our knowledge no numerical software treats them as a data type, with their centre, their family of traces, conditional expectations, inclusions and functional calculus. `torch_vn_algebra` is a PyTorch library that does, for batches of operators on CPUs and GPUs and with automatic differentiation. It fills the gap between computer algebra, which treats algebras as objects but exactly and for small examples, and quantum simulators, which compute at scale but with matrices on a Hilbert space, and it converts operators, states and channels to and from QuTiP and dynamiqs. It is meant for operator algebraists who explore inequalities and extremal problems numerically, for quantum information theorists with variational problems over states and channels under a symmetry, and for physicists who need many small symmetric problems at once. It began as a re-implementation of our earlier NumPy Monte Carlo study of trace inequalities with the Michelson contrast, and this paper documents it. Its design rests on one thesis: keeping the sector structure inside the data type makes computations in the algebra correct by construction and cheap. Haar sampling is exact in every block, trace conventions survive maps between sectors, and all blocks of one size go through one batched kernel. We show that the library reproduces the distributions of the earlier NumPy computation and is about two orders of magnitude faster than its loop formulation for small blocks. More important than speed, the operations are differentiable, which turns the sampling of the earlier study into variational problems over the states, unitaries and channels of the algebra: from block size 4 on, a million Haar samples miss the violation of a trace inequality in every one of 16 test cases, while batched gradient ascent finds the exact supremum in all of them in seconds. We measure each architectural decision against the simple alternative and validate the library against exact results and against QuTiP and dynamiqs. Its generality is illustrated by a noisy XY mixer of the quantum alternating operator ansatz, whose Hamming-weight sectors form the algebra: one batched run gives, for all noise models and rates, the exact effect of post-selection on the weight, and optimising the controls for the post-selected fidelity shows that the measurement of the conserved charge, not the noise-aware design, removes most of the error. Finally, channels between algebras become layers of trainable networks. Learning the noise of the register from measurement counts, models that respect the sectors reach a given accuracy with about fourteen times fewer shots than a dense model, and a model with the known jump operators estimates the rates close to the Cramér–Rao bound; residual networks of channels fit Markovian evolutions but provably not a channel with negative determinant, so their fit error witnesses non-Markovianity.

Program summary

Program title: `torch_vn_algebra`

Developer's repository link: `https://github.com/andre-a-no/von_neumann_type_i`

Licensing provisions: MIT

Programming language: Python (≥ 3.8), PyTorch (≥ 1.13), NumPy (≥ 1.21)

Nature of problem: Numerical experiments in finite-dimensional Type I von Neumann algebras $\bigoplus_c M_{k_c}(\mathbb{C})$ and in the quantum systems with superselection rules or conserved charges that they describe: random operators with prescribed spectra, several trace functionals, states, channels within and between sectors, symmetric tensor products, spin chains, Hamiltonian and Lindblad dynamics with jumps between sectors (noisy circuits with a conserved charge), extremal problems under spectral constraints, inclusions of algebras, symmetry reduction, and quantum Rényi divergences.

Solution method: Batched block-diagonal tensors processed by PyTorch linear algebra on CPUs and GPUs, one call per block size; lazy operators that carry their properties, with caches for spectra; exact Haar sampling; spectral functional calculus with Daleckii–Krein derivatives; Kraus representations; Lanczos and Krylov methods for sparse sector Hamiltonians; sparse jumps between sectors; batched multistart gradient optimisation; run-time and memory estimates before expensive steps; inclusion matrices; randomised Wedderburn decomposition; highest-weight bases and exact Clebsch–Gordan coefficients.

Additional comments: more than 800 unit tests, run in continuous integration; every number, table and figure of this paper is produced by the scripts and examples in the repository and regenerated by `scripts/run_all.sh`. The library is the software used in [9] and in the companion papers [1, 2, 3].

1 Introduction

The algebra as the object. A finite-dimensional von Neumann algebra is, up to $*$-isomorphism, a direct sum of full matrix algebras [33],

$$\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{C}) \quad \text{acting on} \quad \mathcal{H} = \bigoplus_{c=1}^{C} \mathbb{C}^{k_c}, \tag{1}$$

where $M_k(\mathbb{C})$ is the algebra of complex $k \times k$ matrices and the C summands are called sectors. Small as they are, these algebras carry the structure that operator algebra theory studies. Their centre is spanned by the sector projections. Instead of one trace they have a C-parameter family of faithful traces. They have conditional expectations onto subalgebras, inclusions with an index [77, 78], and a Jordan product on their self-adjoint part; their elements have spectral invariants such as the Michelson contrast, a normalised spread of the spectrum (Section 4). Questions about these objects are often explored on examples first: which inequalities between traces of products characterise the trace or the centre [5, 6, 7, 8], and how strongly they fail for functionals that are not traces [9]. Such explorations are usually written as ad hoc matrix code. To our knowledge there is no numerical software in which the algebra (1) is the data type: one declares the sector sizes and then samples, transforms and optimises elements, states and maps, with the centre and the traces built in (Section 3).

From an earlier NumPy study to a library. This library grew out of such an exploration. In [9] and in the first version of this work [4] we studied trace inequalities with the Michelson contrast by Monte Carlo sampling in NumPy. The computation draws many positive operators with prescribed contrast and Haar-random eigenvectors in every block, evaluates a trace functional, and looks at its distribution. Written directly, such code loops over samples and sectors, restates the trace convention

in every script, and has to get the Haar measure right block by block.[1] `torch_vn_algebra` re-implements that modelling as a structured PyTorch library. Section 5 shows that it reproduces the earlier computation: the NumPy formulations and the library sample the same distribution. For small blocks the library is about two orders of magnitude faster than the loop formulation, and for blocks of size 16 about three times faster, and the same code runs on a GPU.

From sampling to optimisation. Speed is the smaller gain. The numerical operations of the library are differentiable (the one exception is the unitary factor of the polar decomposition), including spectral functions (by the Daleckii–Krein formula), channels between sectors and the time steps of Hamiltonian and Lindblad dynamics, so a question asked by sampling can be asked again as a variational problem: maximise a functional over operators with a prescribed spectrum, over states, over unitaries of the algebra or over all channels, by batched gradient ascent with many starts in one call.

Extremal problems in their own right. The variational form is not a detour: in operator algebra, quantum information and physics, many of the central questions are extremal problems. In operator algebra, a trace inequality is settled by its sharp constant [10], and how strongly a characterisation of the trace [5, 6] fails for other functionals is measured by an envelope over unitaries and over spectra with prescribed invariants [9, 1]. The Connes–Størmer entropy of two subalgebras is a supremum over finite partitions of unity [11, 79], and the Pimsner–Popa constant of a conditional expectation is the best constant in an operator inequality, tied to the index [79, 81]. In quantum information, data processing and the parameter ranges in which it holds, channel capacities and minimal output entropies, and optimal recovery are optimisation problems. The convex ones, such as the diamond norm [12] or channel-adapted recovery [13], are semidefinite programs and are best solved by SDP solvers, which also return dual certificates. Many important ones are not convex: the minimal output entropy, whose additivity was disproved with random channels in high dimension [14], suprema over unitaries or over spectra with prescribed contrast, the Rényi and sandwiched quantities outside the ranges in which they are monotone [64, 68], and entropies maximised over partitions. In physics, optimal control of open systems [15, 16], variational state preparation [17] and protocols designed to tolerate noise are optimisation problems over unitaries, states and channels; when the controls and the noise conserve a charge (Section 12), all of their variables lie in an algebra (1).

Sampling is no substitute. Asking such a question by sampling changes the answer, not only the time. The Monte Carlo of [9, 4] saw no violation of Gardner's inequality for blocks of size 16, and we now know why: by concentration of measure, the unitaries near the maximiser have Haar measure that decays rapidly with the dimension. From block size 4 on, a million Haar samples miss the violation in every test case, while gradient ascent finds the exact supremum in all of them, to a relative error of 3×10^{-5} and in seconds (Section 9). The violation envelopes of the companion paper [1], the thresholds of the Rényi divergences in Section 9.1 and the partitions in [2] are all found this way; every reported extremal value is then recomputed at the optimiser in 40- to 50-digit arithmetic (mpmath, in the paper scripts). This gives an independent high-precision evaluation at an explicit point, approximating a lower bound for the supremum; it is not a rigorous numerical certificate with a guaranteed rounding-error bound.

Thesis. The design rests on one thesis: keeping the sector structure inside the data type makes computations in (1) both correct by construction and cheap. By sector structure we mean the blocks,

[1] The first version [4] contained errors of this kind in its samplers; they are corrected and tested here.

the centre, the several traces and the maps between algebras with different sectors. Correctness follows because a random element is sampled block by block from the right measure, a density is always stored with respect to one trace and converted explicitly, and a map between sectors carries its own dual. Cheapness follows because the library knows which blocks have equal size and stacks them into one batched kernel call, knows from property flags which kernel a result needs, and never forms the zero blocks. Section 6 describes the architecture built on this thesis and measures each decision against the simple alternative.

Where the algebra appears in physics. The same algebra is the home of quantum systems with a symmetry. If a group, or a set of conserved quantities, acts unitarily on $\mathcal{H} \cong \bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda}$, with irreducible representations V_λ of multiplicity m_λ, the operators commuting with the action form the algebra $\bigoplus_\lambda \mathbb{1}_{V_\lambda} \otimes M_{m_\lambda}(\mathbb{C}) \cong \bigoplus_\lambda M_{m_\lambda}(\mathbb{C})$. For one conserved charge Q (particle number, magnetisation, Hamming weight) the sectors are the eigenvalues of Q and the centre consists of the functions of Q; a Hamiltonian that conserves Q, the evolution it generates and every observable compatible with a superselection rule for Q lie in this algebra. Examples are superselection rules [34, 35] and decoherence [36], many-body Hamiltonians diagonalised sector by sector [37, 38], and symmetry-resolved entanglement [39, 40]. Others are decoherence-free subspaces and operator-algebra error correction [41, 42, 43], the XY mixers of the quantum alternating operator ansatz [46, 128], random matrices decomposed by symmetry [47, 23], and equivariant layers in machine learning, which lie in such commutants by Schur's lemma [48, 49, 86]. A library built for trace inequalities therefore serves these applications once it has dynamics and maps that move probability between sectors. We added them (spin chains with $U(1)$ and SU(2) sectors, Lindblad dynamics with jumps between sectors, inclusions, a numerical Wedderburn decomposition), and Section 12 uses them for a question that the original NumPy code could not address: symmetry verification in a noisy XY mixer.

Why not one large matrix? $\mathcal{M}$ embeds into $M_D(\mathbb{C})$, $D = \sum_c k_c$, as the block-diagonal matrices, and every matrix package handles that embedding exactly. The question is what it costs and what has to be supplied by hand.

- *Cost.* A product or a decomposition costs $O(D^3)$ time and $O(D^2)$ memory in M_D, but $\sum_c k_c^3$ and $\sum_c k_c^2$ in $\mathcal{M}$: for C equal blocks a factor C^2 in time and C in memory.
- *Measures.* The unitary group of $\mathcal{M}$ is $\prod_c U(k_c)$, for $C \geq 2$ a null set in $U(D)$. A Haar-random element of $U(D)$ is not in $\mathcal{M}$; random elements, random states and gradient steps have to respect the blocks.
- *Traces.* The faithful traces of $\mathcal{M}$ form the family $\sum_c w_c \operatorname{Tr}_c$. Densities, "trace preserving" and the dual of a map between algebras with different sectors depend on the weights, which the embedding forgets.
- *Centre.* Conditioning on a sector, post-selection and sector-resolved quantities are operations with the centre; in M_D they have to be reconstructed from block indices.

For $C = 1$ the library reduces to batched linear algebra in $M_k(\mathbb{C})$, and all its functions accept that case.

What is new and what is re-implemented. None of the numerical methods is new. Haar unitaries come from the QR decomposition of Gaussian matrices with Mezzadri's phase correction [22]. Gradients of matrix functions use the Daleckii–Krein formula [58, 59]. Singular values come from batched SVD or, on GPUs for large blocks, from the Hermitian dilation, holomorphic functions from contour integrals [57], and the Wedderburn decomposition from the randomised algorithms of Murota

and coauthors [83, 85]. Block-sparse storage of charge-conserving tensors is standard in tensor-network libraries [53, 55, 54]. Lazy linear operators that dispatch on their structure are the basis of GPyTorch's `linear_operator` [112, 113] and of CoLA [114]. Our contribution is their combination around the algebra (1) and its conventions: a batched, differentiable data type whose elements know their sector structure, trace convention and spectral properties, with states, channels between algebras, inclusions and dynamics defined on it.

What the paper shows, and for whom. Section 5 reproduces the earlier NumPy study and compares speed. Section 6 explains where the speed comes from, with one measurement per design decision (Table 5), and Table 3 lists seven implementation problems and the sections that resolve them. Monte Carlo estimates agree with exact Haar, Weingarten, Page and purity values within their standard errors (Section 8), and batched optimisation finds extremes that sampling misses (Section 9). Sections 10 and 11 add spin chains, inclusions and symmetry reduction, and Section 12 shows what this generality buys. Section 12.1 turns the noisy register into a design problem: it optimises the controls of a Lindblad evolution for the post-selected fidelity, over a batch of noise models at once. Section 2 places the library among existing software and says for whom it is made and for which tasks another package is the better choice. This paper describes the software used in [9] and in three companion papers [1, 2, 3]; their results are summarised in Section 14 and not repeated.

2 Place in the ecosystem and intended users

The gap. Software that touches finite-dimensional von Neumann algebras sits in two places (Figure 1). Computer-algebra systems treat algebras as objects, with centres, radicals, Wedderburn decompositions and Jordan structures, but in exact arithmetic and for small examples. Simulators, exact-diagonalisation and tensor-network codes compute numerically and at scale, but their object is a matrix or a tensor on a Hilbert space; blocks appear as storage, not as an algebra with a centre, a family of traces and maps between algebras. Generic linear algebra (NumPy, SciPy, PyTorch) is below both, and solvers for semidefinite and manifold problems handle convex or geometric constraints without the structure of (1). What was missing is the numerical counterpart of the first group: a code in which one declares the sector sizes and then samples, transforms, optimises and learns elements, states and channels of the algebra in batches, on CPUs and GPUs, with derivatives and high-precision cross-checks. This is the place of `torch_vn_algebra`. It is built on PyTorch, its test suite checks small cases against exact results, and it exchanges states, operators, channels and Lindblad data with QuTiP and dynamiqs through the module `interop`, so that a computation can move between the corners of the figure.

For whom. We have four groups of users in mind, each with a typical workflow.

Operator algebraists who explore inequalities, extremal problems and invariants numerically before proving them: sharp constants of trace inequalities and their envelopes in terms of the Michelson contrast, the Connes–Størmer entropy as a supremum over partitions, conditional expectations, inclusions and their index, and Jordan products. The library supplies the constraint sets as exact parametrisations and batched multistart search; the paper scripts recompute extremal values in 40- to 50-digit arithmetic (mpmath), so that a numerical observation comes with an independently recomputed candidate value (Sections 9 and 11; the companion papers [1, 2]).

Quantum information theorists with variational problems over states and channels that respect a symmetry or a superselection rule: data processing and its thresholds, recovery maps, diver-

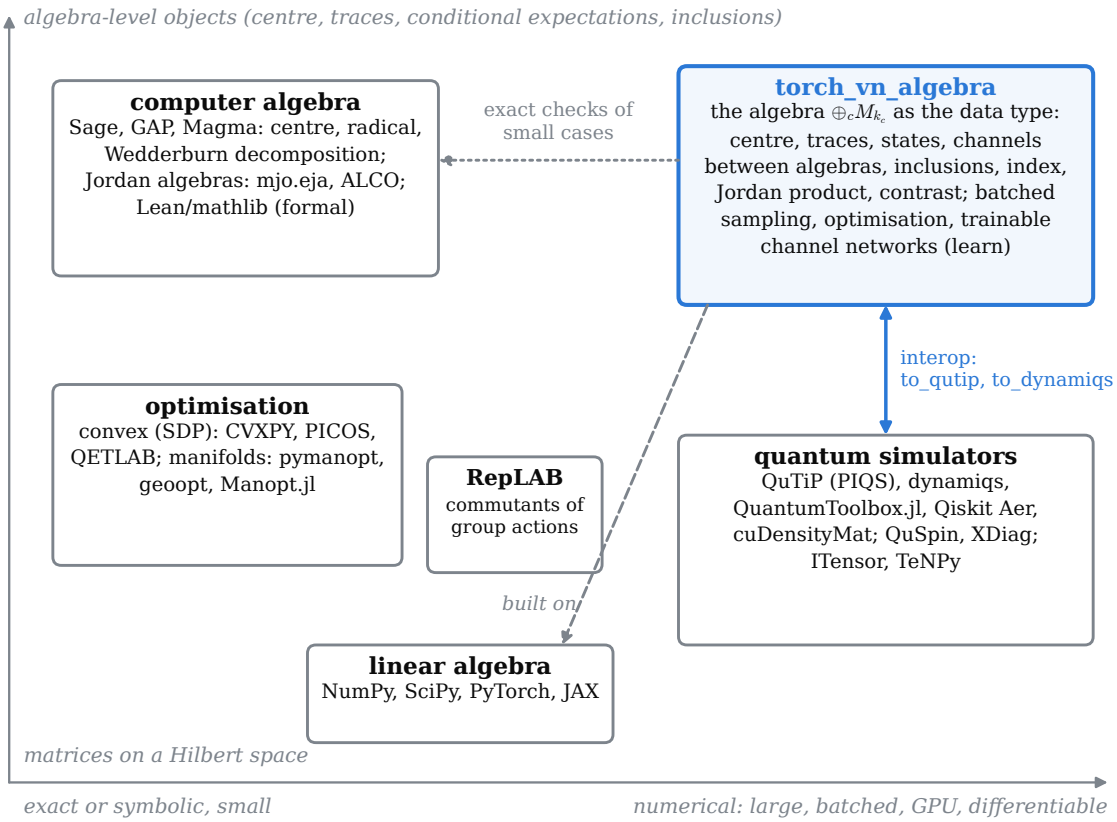


Figure 1: Software for computations with the algebra (1), ordered by two properties: exact and small versus numerical, batched and differentiable (horizontal), and matrices on a Hilbert space versus objects of the algebra itself (vertical). The upper right corner, where the algebra is the data type of a numerical code, was empty; the arrows show what the library builds on, where its results can be checked exactly, and the conversions to the simulators (Section 3).

gences resolved by sector, and learning a channel or a Lindblad generator from measurement data. Channels within and between algebras and Lindblad generators are available as parameters in which every value is physical, and the module `measurement` provides POVMs, sampling, likelihoods, instruments that record their outcome in the centre, and Fisher information (Sections 9.1 and 9).

Physicists with conserved charges who need many small symmetric problems at once rather than one large one: disorder realisations, sweeps over noise models and rates, symmetry verification and post-selection, and noise-aware design of circuits that conserve a charge (Sections 10, 12 and 12.1). A system prepared here can be passed to QuTiP or dynamiqs for further simulation.

Teachers and students of operator algebras and quantum information: the repository contains a course with runnable tutorials in which every construction of the lectures (traces, conditional expectations, the basic construction, channels, Lindblad dynamics) is an object one can sample and test.

A fifth group is emerging: researchers who treat quantum channels as layers of trainable networks. For them the module `learn` provides channel, unitary and Lindblad layers between algebras, residual

blocks, post-selection and feed-forward through the centre as `torch.nn` modules (Section 13).

When to use something else. The library is not a circuit simulator, a tensor-network code or a large-scale exact-diagonalisation package. For the dynamics of one large system, QuTiP, dynamiqs or QuantumToolbox.jl are faster and more complete; for ground states of long chains, QuSpin, XDiag or tensor networks; for pulse-level control, QuTiP-QOC; for convex problems over channels, an SDP solver, which also returns a certificate of optimality; for exact structure of small algebras, a computer-algebra system. The interoperability module exists so that these tools and the library can be used in one workflow.

3 Related software

Software that touches the algebra (1) falls into groups that store matrices or sector blocks for a purpose, usually physics, rather than treating the algebra itself as the object. Table 1 summarises the features relevant here.

Linear algebra. NumPy and SciPy [50, 51], with ARPACK [52] for sparse eigenproblems, handle dense and sparse matrices; stacked arrays give batching, and any block structure is maintained by hand. GPyTorch's `linear_operator` [112, 113] and CoLA [114] go further. They represent structured matrices (diagonal, Kronecker, block-diagonal, low-rank, sums and products) lazily and choose solvers and decompositions by structure, on GPUs and with automatic differentiation. The lazy operators of Section 6.3 follow the same idea. What these packages do not provide is the operator-algebraic layer: the centre, several traces, states, and maps between algebras.

Quantum dynamics. QuTiP [18, 19] and its Julia counterpart QuantumToolbox.jl [94] represent states, operators and superoperators on one Hilbert space and solve Schrödinger, Lindblad and stochastic equations. QuantumToolbox.jl runs on GPUs and supports automatic differentiation; QuTiP 5 does so through optional JAX and CuPy data backends. QuTiP includes PIQS [93], which uses permutational invariance to block-diagonalise the Lindblad dynamics of N two-level systems. PIQS is the closest existing analogue of the sector-wise computations here, specialised to one symmetry. `dynamiqs` [21] provides batched, differentiable Schrödinger and Lindblad solvers on GPUs, built on JAX. Circuit simulators propagate density matrices under noise: Qiskit Aer [20, 95] with noise models on CPUs and GPUs, QuEST [96] on CPUs, GPUs and clusters, and NVIDIA's cuQuantum SDK [97], whose cuDensityMat library [98] integrates master equations for batched operators on GPUs. PennyLane [99] differentiates hybrid quantum-classical circuits, including mixed-state simulation. All of these work on one Hilbert space; a conserved charge, if it is used, is exploited by the user.

Symmetry sectors. Exact-diagonalisation packages build symmetry-reduced bases and sparse sector Hamiltonians: QuSpin [38, 109] for spins, bosons and fermions, and XDiag [110], written in C++ with a Julia interface and distributed over many cores. Tensor-network libraries store charge-conserving tensors block-sparsely: ITensor [53], TeNPy [55], TensorKit.jl [111], and YASTN [54], which, like this library, runs on PyTorch. These packages solve large single problems much better than ours. They do not offer batches of random elements, several traces or maps between algebras.

Algebra software. To our knowledge, no existing numerical library treats finite-dimensional von Neumann algebras $\bigoplus_c M_{k_c}$ as a data type with a centre, traces and states, conditional expectations,

inclusions with their index, and functional calculus. Computer-algebra systems provide exact structure theory for small algebras. Examples are Euclidean Jordan algebras with spectral and Peirce decompositions in Orlitzky's SageMath package [115], the Jordan algebras of SageMath, the GAP package ALCO [116], an R package [117], and centres, radicals and algebras with involution in SageMath, GAP and Magma [118]. Numerical work block-diagonalises matrix $*$-algebras as a preprocessing step for semidefinite programming [83, 84, 119, 120]. In the same vein, RepLAB [121] decomposes representations of compact groups by sampling their commutant, and SymbolicWedderburn.jl [122] computes Wedderburn decompositions of group-symmetric problems exactly. RepLAB's commutant object is the closest numerical analogue; it has no centre or state interface, inclusions, index or functional calculus. Conic solvers hard-code the Jordan products of specific symmetric cones [123, 124]. Subfactor software is combinatorial, working with principal graphs and planar algebras [125, 126]. Proof assistants formalise von Neumann algebras and continuous functional calculus without computing with them [127]. `torch_vn_algebra` is numerical, batched, GPU-capable and differentiable, which is what Monte Carlo sampling and optimisation over the algebra need; it takes the block structure as input, and its own Wedderburn decomposition (Section 11) can also supply it.

Optimisation. Several packages differentiate quantum dynamics. QuTiP 5 does so through its JAX data layer, QuTiP-JAX, which gives gradients of Lindblad and Monte Carlo solutions with respect to Hamiltonian and counting-field parameters [19]. `dynamiqs` differentiates its Schrödinger, Lindblad and stochastic solvers with respect to arbitrary parameters and batches over Hamiltonians, initial states and jump operators [21]. QuantumToolbox.jl has experimental forward- and reverse-mode differentiation of its master-equation solver [94], Qiskit Dynamics uses JAX for differentiable simulation and control optimisation [100], and PennyLane differentiates noisy circuits on its mixed-state simulator by backpropagation [99]. All of these differentiate a trajectory in the full Hilbert space with respect to physical parameters. QuTiP-QOC optimises control pulses with GRAPE, CRAB, GOAT and the JAX-based JOPT, also for open systems through their Liouvillians [19]. Convex problems over states and channels, such as diamond norms [12] and channel-adapted recovery [13], are semidefinite programs. The modelling languages CVXPY [101] and PICOS [102], both with complex matrix variables, and the MATLAB toolbox QETLAB [103] solve them to the global optimum with a dual certificate. Riemannian optimisation libraries supply manifolds: Pymanopt [104] includes the unitary group and its powers $U(n)^m$, geoopt [105] and McTorch [106] add Stiefel manifolds to PyTorch, and Manopt.jl [107] works on the real and complex manifolds of Manifolds.jl [108].

What `torch_vn_algebra` adds lies between these groups. Its constraint sets are those of the algebra, parametrised exactly and batched: the unitary group $\prod_c U(k_c)$ with blocks of different sizes, positive and self-adjoint elements with a prescribed spectrum or Michelson contrast, channels within one algebra and between algebras with different sectors (`StinespringParam`) and Lindblad generators (`LindbladParam`); states and partitions of unity are parametrised in the paper scripts (Sections 9.1 and 14). Objectives can use the centre, since sector probabilities, post-selection on a charge and conditioning on a sector are operations of the data type. Independent starts, parameter values and models are elements of one batch, so an envelope over a grid, or a threshold scan over all α of Section 9.1, is one call. The paper scripts recompute reported extremal values at the optimiser in 40- to 50-digit arithmetic. The exact value at a feasible point is a lower bound for the supremum; its computed approximation has no guaranteed error enclosure, and local search supplies no global upper bound. Other packages are better in three respects. A convex problem belongs to an SDP solver, which returns the global optimum with a certificate where our local search returns a lower bound. A single large system belongs to the simulators above. Pulse-level control, with waveform constraints and optimisation of the evolution time, is the purpose of QuTiP-QOC. The library is meant for problems that are non-convex, live in an algebra (1), and come in many instances.

	NumPy/SciPy	QuTiP 5 (PIQS)	QuantumToolbox.jl	`dynamiqs`	Qiskit Aer, QuEST	cuDensityMat	QuSpin, XDiag	ITensor, TeNPy, YASTN	lin. operator, CoLA	this work
GPU execution	–	(✓)	✓	✓	✓	✓	–	(✓)	✓	✓
batched over many problems	(✓)	–	(✓)	✓	–	✓	–	–	✓	✓
automatic differentiation	–	(✓)	✓	✓	–	(✓)	–	(✓)	✓	✓
noisy (Lindblad) dynamics	–	✓	✓	✓	✓	✓	–	–	–	✓
symmetry sectors	–	(✓)	–	–	–	–	✓	✓	(✓)	✓
centre, several traces, cond. expectations	–	–	–	–	–	–	–	–	–	✓
maps between algebras of different sectors	–	(✓)	–	–	–	–	–	–	–	✓
Haar/COE/CSE and prescribed spectra per sector	(✓)	(✓)	–	–	–	–	–	–	–	✓
optimisation over the constraint sets of $\mathcal{M}$	–	–	–	–	–	–	–	–	–	✓
large single systems	✓	✓	✓	✓	✓	✓	✓	✓	✓	(✓)

Table 1: Features relevant to computations in the algebra (1), to the best of our knowledge in October 2026 (✓: supported; (✓): partly, through optional backends or plug-ins, for one symmetry (PIQS), or with user code; –: not supported). QuTiP 5 uses GPUs and automatic differentiation through its JAX and CuPy backends; PIQS handles the sectors of permutational symmetry and the maps between them. Qiskit Aer runs density-matrix simulations with noise models on GPUs, QuEST on GPUs and clusters. Optimisation over the constraint sets of $\mathcal{M}$ means unitaries of $\prod_c U(k_c)$ and elements with prescribed spectra as parameters; pulse optimisation (QuTiP-QOC), SDP solvers and manifold libraries are compared in the text. The last row is where the specialised packages are the better choice.

Working with QuTiP and dynamiqs. The library is meant to complement the simulators, not to replace them. The module `interop` converts operators, states and channels to and from `qutip.Qobj` and dynamiqs arrays, and Lindblad jump operators to collapse operators; for a spin chain it places the sectors in the computational basis with QuTiP's conventions. A map between algebras is a map into the algebra, so it is converted together with the pinching onto both input and output sectors: a general `InterSectorChannel` has dense Kraus operators $P_d K_i P_c$, with independent Kraus indices for each input–output sector pair. This preserves the block-diagonal Lindblad loss term as well as the action on states. A `SectorJump` instead exports its coherent physical jump operator. A dense matrix outside the algebra is rejected on the way back. Through this module the Lindblad solvers of Section 12 are checked against `qutip.mesolve` and `dynamiqs.mesolve` in the test suite: on a four-qubit register with damping and dephasing the exact stepper agrees with both to better than 10^{-9}, and the CPTP splitting used in Section 12 converges to them at second order. A computation can move to whichever package suits its next step, for example to QuTiP for a large single system after the sector structure has been used to set it up.

4 Mathematical setting

Algebra and traces. Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$ act on $\mathcal{H} = \bigoplus_c \mathbb{C}^{k_c}$; an element $A \in \mathcal{M}$ is block diagonal, $A = \bigoplus_c A_c$. Every trace on $\mathcal{M}$ is $\mathrm{tr}_w(A) = \sum_c w_c \operatorname{Tr} A_c$ with weights $w_c \geq 0$, faithful iff all

$w_c > 0$. The library implements three of them,

$$\mathrm{Tr}_{\mathrm{blunt}}:\ w_c = 1, \qquad \mathrm{Tr}_{\mathrm{norm}}:\ w_c = \tfrac{1}{k_c}, \qquad \tau_{\mathrm{vN}}:\ w_c = \tfrac{1}{Ck_c}.$$

$\mathrm{Tr}_{\mathrm{blunt}}$ is the plain trace of the block-diagonal matrix; we call it "blunt" because it ignores the block sizes. $\mathrm{Tr}_{\mathrm{norm}}$ normalises each block, and τ_{vN} is the faithful tracial state that gives every sector the same weight. Empty sectors ($k_c = 0$) are allowed, carry no weight, and are not counted in C for τ_{vN}. The library also accepts nominal factor sizes $n_c \geq k_c$ and then works in the corner $p\mathcal{M}_0 p \cong \bigoplus_c M_{k_c}$ of $\mathcal{M}_0 = \bigoplus_c M_{n_c}$ (p_c of rank k_c); all traces refer to the corner, and in most applications $k_c = n_c$. For $A \in \mathcal{M}$, $|A| = (A^*A)^{1/2}$. For $A \geq 0$, $A \neq 0$, the *Michelson contrast* is $\Delta(A) = (\lambda_{\max} - \lambda_{\min})/(\lambda_{\max} + \lambda_{\min}) \in [0, 1]$, with the extreme eigenvalues taken over all sectors; it is the operator version of the fringe visibility of optics, and $\Delta(A) = 0$ iff A is a multiple of the identity (the library returns $\Delta(0) = 0$).

States. A state ω on $\mathcal{M}$ is represented by its density $\rho \geq 0$ with respect to $\mathrm{Tr}_{\mathrm{blunt}}$, $\omega(A) = \mathrm{Tr}(\rho A)$, $\sum_c \mathrm{Tr}\,\rho_c = 1$. Its restriction to the centre is the classical distribution $p_c = \mathrm{Tr}\,\rho_c$ over sectors. With respect to tr_w the same state has the density $\rho_c^{(w)} = \rho_c / w_c$; the library converts between conventions explicitly.

Maps between sectors. Let $\mathcal{M}_{\mathrm{in}} = \bigoplus_c M_{k_c}$ and $\mathcal{M}_{\mathrm{out}} = \bigoplus_d M_{m_d}$. Every normal completely positive map $\Phi : \mathcal{M}_{\mathrm{in}} \to \mathcal{M}_{\mathrm{out}}$ (Schrödinger picture) is a sum of completely positive maps between blocks,

$$\Phi(\rho)_d = \sum_c \sum_i K_i^{dc}\, \rho_c\, (K_i^{dc})^*, \qquad K_i^{dc} : \mathbb{C}^{k_c} \to \mathbb{C}^{m_d}.$$

It preserves $\mathrm{Tr}_{\mathrm{blunt}}$ iff $\sum_{d,i} (K_i^{dc})^* K_i^{dc} = 1_c$ for all c. For an input state with $\mathrm{Tr}\,\rho_c > 0$, $T_{dc} = \mathrm{Tr}\,\Phi_{dc}(\rho_c)/\mathrm{Tr}\,\rho_c$ are the transition probabilities between sectors. They do not depend on ρ iff every $\sum_i (K_i^{dc})^* K_i^{dc}$ is a multiple of 1_c; by default the library uses the maximally mixed state of each input sector. The dual with respect to tr_w on both sides is $\Phi^{*,w}(A)_c = \sum_{d,i} (w_d^{\mathrm{out}} / w_c^{\mathrm{in}}) (K_i^{dc})^* A_d K_i^{dc}$. For sector-preserving maps on one algebra the weights cancel and all three traces give the same dual.

Tensor products and fusion of sectors. $\left(\bigoplus_c M_{k_c}\right) \otimes \left(\bigoplus_d M_{m_d}\right) = \bigoplus_{(c,d)} M_{k_c m_d}$ has sectors labelled by pairs. If the sectors carry additive charges q_c, q'_d, the operators that conserve only the *total* charge form the larger algebra

$$\mathcal{M}_1 \hat{\otimes} \mathcal{M}_2 = \bigoplus_q M_{n_q}, \qquad n_q = \sum_{q_c + q'_d = q} k_c m_d.$$

It contains $\mathcal{M}_1 \otimes \mathcal{M}_2$ and also operators such as $S_i^+ S_j^-$ that move charge between the factors. If distinct sectors of $\mathcal{M}_1$ carry distinct charges, the partial trace $(\mathrm{Tr}_2\, X)_c = \sum_d \mathrm{Tr}_{m_d} X_{(c,d),(c,d)}$ still lands in $\mathcal{M}_1$. In both cases it is a map between sectors, and its dual with respect to $\mathrm{Tr}_{\mathrm{blunt}}$ is $A \mapsto A \otimes 1$.

Spin chains and dynamics. For L spins 1/2 with conserved S^z_{tot}, iterating the fused product of single sites $M_1 \oplus M_1$ (charges $0, 1$) gives $\mathcal{M} = \bigoplus_{N=0}^{L} M_{\binom{L}{N}}$, N the number of up spins. It contains every S^z_{tot}-conserving Hamiltonian, e.g. the XXZ chain $H = \sum_{\langle ij \rangle} \left[\frac{J}{2}(S_i^+ S_j^- + S_i^- S_j^+) + J\Delta_z S_i^z S_j^z\right] + \sum_i h_i S_i^z$. For $H = H^* \in \mathcal{M}$ and jump operators $L_j \in \mathcal{M}$ the Lindblad equation $\dot{\rho} = -i[H, \rho] + \sum_j \gamma_j (L_j \rho L_j^* - \frac{1}{2}\{L_j^* L_j, \rho\})$ preserves every sector, so the sector probabilities p_c are conserved. Jumps between sectors, such as particle loss, are maps between sectors as above.

```python
import torch
from torch_vn_algebra import TypeIAlgebra

alg = TypeIAlgebra([4, 4, 8], [4, 4, 8], complex_valued=False,  # M_4 + M_4 + M_8, real,
                   precision='double')                          # as in Table baselines
B = 100_000                                    # Monte Carlo samples: one batch

def spectrum(dim, delta=0.6):                  # lambda_max = 1, Michelson contrast delta
    lo = (1 - delta) / (1 + delta)
    lam = lo + (1 - lo) * torch.rand(B, dim, dtype=torch.float64)
    lam[:, 0], lam[:, 1] = 1.0, lo
    return lam

X = alg.operator_from_eigenvalues(spectrum, batch_size=B, force_positive=True)
Y = alg.operator_from_eigenvalues(spectrum, batch_size=B, force_positive=True)
U = alg.random_unitary_operator(B)             # Haar-orthogonal in every block
z = (X @ U @ Y).trace.abs() - (X @ Y).trace    # z = |Tr(XUY)| - Tr(XY)
```

Listing 1: The Monte Carlo of (2) in $M_4 \oplus M_4 \oplus M_8$ over $\mathbb{R}$: 10^5 samples in one batch. `operator_from_eigenvalues` draws independent Haar eigenvectors in every block, `trace` is $\mathrm{Tr}_{\mathrm{blunt}}$.

5 Reproducing the earlier NumPy study

The Monte Carlo experiments of [9] (Sect. 5, a single block $M_n(\mathbb{R})$, $n \le 5$) and of [4] (several sectors) sample trace functionals such as

$$z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY), \qquad X, Y \ge 0, \quad U \in \mathcal{M} \text{ unitary}, \tag{2}$$

with $\mathrm{Tr} = \mathrm{Tr}_{\mathrm{blunt}}$, the operators normalised to $\lambda_{\max} = 1$ and their contrasts as parameters ("Exp. 1" in Section 9). Listing 1 is the complete library version: it declares a real algebra, samples 10^5 pairs with prescribed contrast and Haar-orthogonal eigenvectors in every block, and evaluates z for the whole batch.

To check that nothing was lost in the re-implementation, the same computation was written three ways (`scripts/baselines.py`). The loop formulation draws one sample and one sector at a time with `scipy.stats.ortho_group`, as code written directly from the definition does. The hand-vectorised NumPy version stacks QR decompositions with the phase correction. The library version is Listing 1. The three versions sample the same distribution: their means of z differ by at most 1.2 standard errors.

Table 2 shows the times. Against the loop formulation the library gains about two orders of magnitude for blocks of size 2, where the Python overhead dominates, and about a factor of three for 16 or 32 blocks of size 16, where both spend their time in the same LAPACK routines. Against carefully vectorised NumPy the library is not systematically faster on a CPU, because both call the same kernels. Its value there is that the vectorised, correctly sampled, block-aware version is the default rather than an expert rewrite, and that the same code runs on a GPU and can be differentiated. GPU timings depend on the precision: the consumer GPU of this run executes FP64 at 1/64 of its FP32 rate, and in double precision it does not beat the CPU on this workload. Section 6 explains where the speed comes from, and Section 9 shows that the batched optimisation finds extremes of (2) that no amount of sampling reaches.

C	k	NumPy loop	NumPy vectorised	library (CPU)	library (GPU)
1	2	160	2.49	1.50	124
16	16	2852	690	884	2000
32	16	5702	1380	1688	4005

Table 2: Time per Monte Carlo sample [μs] of (2) for C real sectors of size k, in double precision unless stated (median of up to five runs after a warm-up; full run on NVIDIA GeForce RTX 5090).

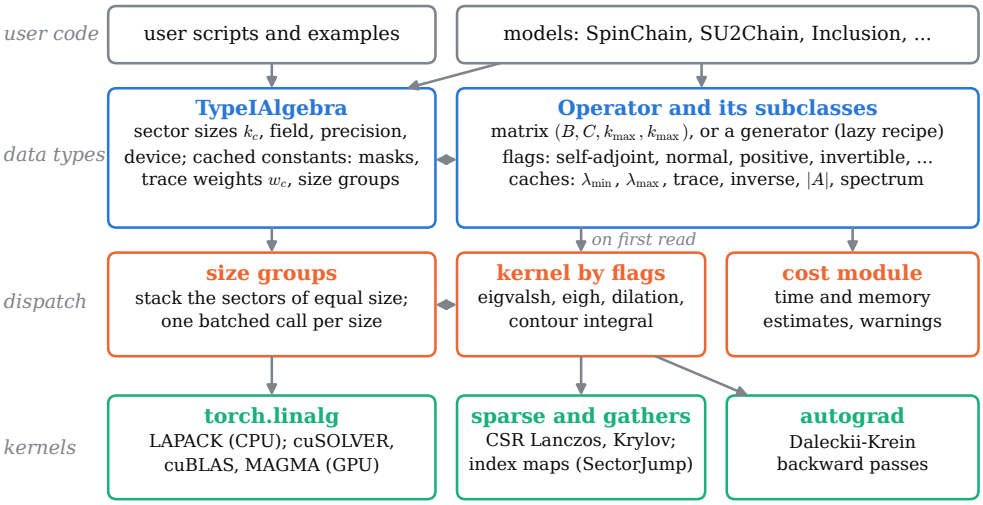


Figure 2: Architecture. User code and models create operators in a `TypeIAlgebra`; operators are matrices or lazy generators that carry property flags and caches. When a result is read, the dispatcher groups sectors of equal size and chooses the kernel from the flags; the cost module estimates time and memory first. The work ends in batched `torch.linalg` calls, sparse products or index gathers, and is differentiated by autograd with custom backward passes.

6 Design and implementation

6.1 Software stack and workflow

The library is pure Python on top of PyTorch [92] ($\geq$ 1.13) and NumPy [50] (Figure 2). Dense linear algebra goes through `torch.linalg`, i.e. LAPACK (MKL or OpenBLAS) on CPUs and cuSOLVER, cuBLAS and MAGMA on NVIDIA GPUs; sparse products use PyTorch's CSR tensors; automatic differentiation is PyTorch's reverse mode. The validation and paper scripts also use SciPy, pandas, Matplotlib and mpmath. The test suite runs in continuous integration on Python 3.8–3.13 and with the oldest supported versions (Python 3.8, PyTorch 1.13, NumPy 1.21). The paper scripts record the device, the matmul precision, the PyTorch, CUDA and cuDNN versions and the commit in a JSON file next to their tables; files written by earlier versions of a script lack some of these fields.

A computation proceeds in four steps. (1) An algebra is declared by its sector sizes, and optionally sector charges, number field, precision and device (`TypeIAlgebra`, or a model such as `SpinChain` that creates it). (2) Objects are created in it: random operators with prescribed spectra, states, channels, Hamiltonians, or differentiable parametrisations. (3) They are combined lazily: products, functional calculus, channels, tensor products, time evolution. (4) Functionals are evaluated for the whole batch at once and reduced to statistics, or fed into an optimiser that updates the parameters of step (2).

problem	solution in the library	section
(i) blocks of different size on batched kernels	padded $(B, C, k_{\max}, k_{\max})$ layout; one batched call per distinct block size; single sectors and sparse sector Hamiltonians where padding would dominate	6.2, 6.4
(ii) conventions between sectors	states stored as $\mathrm{Tr}_{\mathrm{blunt}}$-densities with explicit conversions; duals computed with the weights of the chosen trace	4
(iii) exact random ensembles	Mezzadri's QR algorithm with phase correction; $W^T W$ and $W^R W$ for COE/CSE; checked against Weingarten moments and spacing statistics	6.5, 8
(iv) singular gradients at degenerate spectra	Daleckii–Krein backward passes; parametrisations that build operators from their spectra	6.5
(v) extremes missed by sampling	batched multistart optimisation with exact contrast constraints	9
(vi) time and memory budgets	probing of batched kernels with time estimates; memory checked against the free device memory	6.7
(vii) field and precision	two switches per algebra, conversions, measured trade-offs	6.6

Table 3: Implementation problems and where they are resolved.

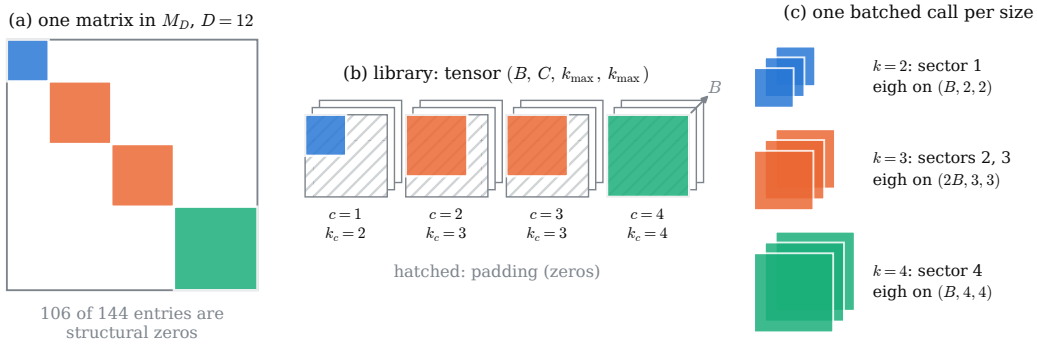


Figure 3: Data layout for $\mathcal{M} = M_2 \oplus M_3 \oplus M_3 \oplus M_4$. (a) As one block-diagonal matrix most entries are structural zeros. (b) The library stores a batch of B operators as a $(B, C, k_{\max}, k_{\max})$ tensor, each block in the top-left corner of its slice. (c) Decompositions stack the blocks of equal size, so the two sectors of size 3 and all samples go into one batched call.

Every step keeps the batch dimension, so the same code serves one operator and 10^5 samples. Table 3 lists the implementation problems this raises.

6.2 Data layout and objects

An operator is a tensor of shape $(B, C, k_{\max}, k_{\max})$: B samples, C sectors, block c in the top-left $k_c \times k_c$ corner and zeros elsewhere (Figure 3). Linear algebra is applied to the batch of active blocks, so one call to a batched LAPACK or cuSOLVER routine serves all samples. Padding costs memory when the sector sizes differ strongly: for a chain of 14 spins, 77 % of a padded operator is zeros. The library warns in that case, and `SpinChain` lets one work in single sectors $M_{\binom{L}{N}}$ without padding, which is what dense diagonalisation needs anyway. `Operator`, with the subclass `DensityMatrix` for states, `Channel` (sector preserving) and `InterSectorChannel` (between sectors or algebras) are thin wrappers around such tensors; `SectorJump` is a sparse jump operator for the Lindblad solvers (Section 6.4). Table 4 lists the modules.

module	content
`algebra`	`TypeIAlgebra`, `Operator`: random operators, unitary ensembles, traces, functional calculus
`states`	`DensityMatrix`, random and Gibbs states, Lüders rule, entropies, fidelity
`channels`	`Channel`, `InterSectorChannel`, `SectorJump`, Choi matrices, standard and random channels
`composite`	plain and fused (charge-conserving) tensor products, `kron`, partial traces
`chains`, `krylov`	`SpinChain`: $U(1)$ and momentum sectors, XXZ Hamiltonians with batched disorder, jumps between sectors; sparse sector Hamiltonians, batched Lanczos and Krylov
`dynamics`	exact Schrödinger/von Neumann evolution, RK4, Lindblad equation (RK4, completely positive splitting, $e^{t\mathcal{L}}$)
`functional`	functional calculus: self-adjoint, normal, singular values, holomorphic (contour); spectral cache
`optimize`, `cost`	parametrisations with prescribed contrast, channels (`StinespringParam`) and Lindblad generators (`LindbladParam`); batched multistart `extremize` and `fit`; run-time and memory estimates
`measurement`	POVMs (sectors, bases, rotated and randomised settings), sampling, likelihoods, instruments recording the outcome in the centre, Fisher information; projection-valued measures with values (`PVM`)
`anyons`	`AnyonChain`: fusion paths of $SU(2)_k$ anyons, Temperley–Lieb generators, the tower $TL_{L-1} \subset TL_L$ as `Inclusion`, sparse Hamiltonians (Section 11.2)
`direct_integral`	`DirectIntegral`: a measure space by nodes and weights (product rules, torus, Monte Carlo, partitions); fields, trace, normal states, central projections (Section 11.1)
`projections`	projection lattice ($P \vee Q$, $P \wedge Q$, ranks, Haar random projections); measures on projections (states by Gleason's theorem, frame functions on 2×2 blocks, dispersion-free measures); spectral measures of self-adjoint operators
`inclusions`	`Inclusion`: inclusion matrices, conditional expectations, Markov trace, Jones, Watatani and Pimsner–Popa indices, basic construction, spin-chain tower
`decompose`, `su2`	numerical Wedderburn decomposition, commutants, real/complex/quaternionic types; `SU2Chain`: SU(2) multiplets, Clebsch–Gordan coefficients, Wigner–Eckart theorem
`divergences`, `jordan`	Petz and sandwiched Rényi divergences, recovery maps, L^p norms; Jordan product, quadratic representation, spin factors
`permutational`	N two-level systems with permutation symmetry: total-spin sectors, collective operators, Dicke states, local jumps between sectors (`PermutationalLindbladian`)
`interop`	conversion to and from dense matrices, QuTiP and dynamiqs (chain layout, pinching of maps)
`learn`	channels as trainable networks: channel, unitary and Lindblad layers, residual blocks, post-selection, feed-forward through the centre; losses (Section 13)

Table 4: Modules of `torch_vn_algebra`.

6.3 Lazy evaluation and caches

Operators are recipes until used. An `Operator` holds either its matrix or a *generator*, a closure that computes it. Sums, products, adjoints, functions $f(A)$, channel images $\Phi(\rho)$ and maps between sectors return new operators that hold only a generator. The matrix is formed on the first access to `.matrix`: the generator runs, its result is stored, and the generator is dropped so that it no longer keeps the parent matrices alive. A long expression therefore costs exactly the products needed for the requested result, and branches that are never read are never computed. The batch size of a lazy operator is known without computing it, so shapes and memory estimates are available before any work is done. Laziness defers and skips work but does not fuse kernels. Where fusion matters, the library uses direct formulas instead: $\mathrm{Tr}(\rho A) = \sum_{ij} \rho_{ij} A_{ji}$ in $O(k^2)$ per block without forming ρA, and the sector probabilities of a channel image as one contraction $\sum K_{mi} \rho_{ij} \overline{K_{mj}}$ without forming the

image.

Properties travel with the recipe. Every operator carries flags (self-adjoint, normal, positive, invertible, projection) that are known, unknown, or checked. Operations set the flags that hold mathematically: sums of self-adjoint operators are self-adjoint, products of invertible operators invertible, and an operator built from its spectrum is self-adjoint, and positive if the spectrum is. Flags select the kernel. A self-adjoint operator uses the Hermitian eigensolver for its norm, trace norm, inverse and $|A|$, never an SVD, and $\lambda_{\max}$ of an operator built from its spectrum is read from the spectrum. An unknown flag is computed once, on demand, and stored. Flags passed by the user are trusted on eager operators and validated when a lazy operator is materialised, with one host synchronisation for all checks.

Caches. The extreme eigenvalues (from one eigenvalue call, without eigenvectors), the trace, the inverse and $|A|$ are cached on the operator. The eigendecomposition itself is kept only on request (`cache_spectrum()`, or the context `caching()`); afterwards every function of the operator, for instance e^{-itH} for many t, costs two matrix products. The cache is keyed by the identity and version counter of the stored tensor, so an in-place change invalidates it. The algebra caches its own constants on the device (block sizes, masks, identities, trace weights, index tensors of the size groups), so no Python list is copied to the GPU inside a loop.

6.4 Where the speed comes from

Three principles account for the speed, and Table 5 measures several of the decisions behind them by computing the same result with and without each. The alternatives are deliberately the simple ones a direct implementation would use, so the factors measure the value of each decision, not a comparison with other software.

The first principle is to give every kernel as much independent work as possible. Every operation acts on the whole batch, so a Monte Carlo study of 10^5 samples is one tensor program, not 10^5 calls. Decompositions are applied to the stacked active blocks of all sectors of equal size, one batched call per distinct size (Figure 3c); contiguous sectors are selected as views, and an unpadded algebra needs no scatter.

The second principle is to compute only what is read. One batched Hermitian eigensolver serves spectra, $f(A)$ and normal operators, and on GPUs also $|A|$, singular values and polar decompositions of blocks larger than 32 (Section 6.5); it is fast and, unlike batched SVD on GPUs, not limited in block size. Where only the spectrum is needed (extreme eigenvalues, contrasts, entropies), eigenvectors are not formed. Where the spectrum is known from the construction, no decomposition is run at all, and a cached eigendecomposition turns many functions of one operator into products.

The third principle is to keep the device busy and the memory small. Inner loops do not call `.item()` or branch on device values, so checks synchronise once per call, and the Lindblad steppers prepare constant propagators once per step size. Sparse structure is used where it exists. Sector Hamiltonians of spin chains are CSR matrices for batched Lanczos and Krylov solvers (Section 10). Jumps that map basis vectors to multiples of basis vectors ($S_i^{\pm}$, S_i^z, particle loss) are stored as `SectorJump`, with a target sector per sector and a target position and coefficient per basis vector. The Lindblad solvers then apply $L\rho L^*$ as a weighted permutation of the entries of ρ, L^*ZL as a gather and L^*L as a diagonal, in $O(\sum_c k_c^2)$ operations against $O(C^2 k_{\max}^3)$ for a dense map between sectors. A batch of coefficients gives one jump per sample, so a sweep over noise rates is one batched evolution.

mechanism	computation	library [ms]	without [ms]	speed-up
known spectrum	$\lambda_{\max}$, 256 operators in $M_{32}^{\oplus 8}$	2.25	28.70	13
eigenvalues only	spectra of the same operators	28.27	93.60	3.3
spectral cache	e^{-itA} for 50 values of t	3019.57	11717.58	3.9
size groups	spectra, 256 operators in $M_8^{\oplus 64}$	19.69	73.53	3.7
trace of a product	$\mathrm{Tr}(\rho A)$, 32 pairs in $M_{256}^{\oplus 2}$	16.62	41.75	2.5
sparse jumps	Lindblad, 8 qubits, S_i^- and S_i^z	354.63	1010.80	2.9

Table 5: The same result with and without one design decision (best of several runs, double precision, CPU with 4 threads; `scripts/architecture_bench.py`). "Without": eigenvalues by `eigvalsh` instead of the known spectrum; `eigh` instead of `eigvalsh`; one decomposition per time instead of the spectral cache; a Python loop over sectors; forming ρA; dense jumps between sectors. For the last row the eight jumps S_i^- take 48.4 MiB as dense maps and 0.12 MiB as `SectorJump`s, and the two evolutions agree to 4×10^{-15}; its time is dominated by the dense Hamiltonian, which both versions share.

6.5 Random unitaries, functional calculus and derivatives

Random unitaries. Haar unitaries on $U(n)$ and $O(n)$ come from the QR decomposition of a matrix with i.i.d. complex (real) Gaussian entries after fixing the phases (signs) of $\operatorname{diag} R$ [22]; $SU(n)$ and $SO(n)$ follow by multiplying with $\det^{-1/n}$ or flipping the sign of one column. The circular orthogonal and symplectic ensembles are $W^T W$ and $W^R W$ with W Haar on $U(n)$ (n even for $W^R W$, W^R the quaternion dual) [23]. Section 8 checks all of them.

Functional calculus. The module `functional` provides $f(A)$ at four levels of generality, each with batched kernels and one call per block size [56]. (1) For self-adjoint A, $f(A) = V f(\Lambda) V^*$ from the Hermitian eigensolver. (2) For normal $A = H + iK$, H and K commute, and the eigenvectors of $H + \alpha K$ diagonalise A for all but finitely many α. The library tries a few irrational values, refines nearly tied eigenvectors with a second coefficient and corrects the remaining residual to first order, so normal operators also need only the Hermitian eigensolver. (3) Singular values, $|A|$, the pseudoinverse and the polar decomposition come from batched SVD, except on GPUs for blocks larger than 32, where batched SVD falls back to a loop; there they come from the Hermitian eigensolver of the dilation $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$, whose eigenvalues are $\pm s_i$. (4) For arbitrary A and f holomorphic near the spectrum, $f(A) = (2\pi i)^{-1} \oint f(z)(z - A)^{-1} dz$ is evaluated by the trapezoidal rule on a circle around the spectrum [57], i.e. by batched LU solves, doubling the nodes until two results agree; a result whose error estimate stays above the square root of the tolerance raises an error, and one between the two triggers a warning.

Differentiation. All numerical operations are differentiable, except the unitary factor of the polar decomposition, which is returned without gradient. Gradients propagated naively through eigenvectors contain factors $1/(\lambda_i - \lambda_j)$ and fail at degenerate spectra, which are typical here (projections, symmetric and extremal operators). The functional calculus therefore has its own backward pass: by the Daleckii–Krein formula [58, 59], the derivative of f at self-adjoint A is $dF = V\,(L \circ V^* dA\, V)\, V^*$, with the Loewner matrix $L_{ij} = (f(\lambda_i) - f(\lambda_j))/(\lambda_i - \lambda_j)$ of divided differences, replaced by f' for coinciding eigenvalues, which is finite whenever f is differentiable on the spectrum. Backward passes through the eigenvectors stay finite at degenerate spectra only for losses that do not mix a degenerate eigenspace [60], and they then miss the entries $f'(\lambda)$ inside the degenerate blocks. The same backward pass serves $\sqrt{A}$, $\log A$, A^p, $|A|$ (as $\sqrt{A^* A}$, with $L_{ij} = 1/(s_i + s_j)$) and inverses of self-

adjoint operators (the pseudo-inverse of a general operator has a constant-rank backward pass of its own); normal A use complex divided differences, and the contour integral is differentiated through its solves. For `apply_function(per_block=True)`, a smooth permutation-equivariant function may couple the eigenvalues of one block. Its full Jacobian J supplies the diagonal eigenvalue sensitivities, and the off-diagonal derivative at a repeated eigenvalue is $J_{ii} - J_{ij}$; computing J takes k reverse passes and $O(k^2)$ storage per block. Both the elementwise and coupled paths retain the dependence of function parameters on the spectrum, including mixed second derivatives. Second derivatives use a differentiable eigendecomposition and require distinct eigenvalues.

6.6 Number field and precision

Every algebra has two switches, `complex_valued` and `precision` (`'single'` or `'double'`), giving `float32`, `float64`, `complex64` and `complex128`; everything built on the algebra inherits them. `alg.like(...)` creates the same sector structure with other numerics and `op.cast(alg2)` moves an operator there, for instance to explore in single precision and re-evaluate the interesting cases in double precision.

The field is part of the model. Real matrices change the symmetry group and the random ensembles, not only the speed. Haar matrices of a real algebra are orthogonal ($\mathbb{E}|U_{11}|^4 = 3/(n(n+2))$ instead of $2/(n(n+1))$, Table 9). Real symmetric and complex Hermitian random matrices have mean spacing ratios 0.5307 (GOE) and 0.5996 (GUE) [32, Table I], against 0.5313 and 0.5993 in our samples. Real arithmetic loses nothing when the problem is real, e.g. for ground states of Hamiltonians without complex hoppings, or for the supremum of $|\operatorname{Tr}(XUY)|$ over unitaries for real $X, Y \geq 0$, which is attained by an orthogonal matrix. Complex matrices are required for e^{-iHt} and Lindblad dynamics with a Hamiltonian (the library refuses to run them in a real algebra), for Haar unitaries, CUE/CSE and random pure states, and for systems without an antiunitary symmetry squaring to $+1$.

Precision. Single precision resolves about 10^{-7} relative to the largest entry, enough for Monte Carlo statistics and exploratory optimisation. Double precision is needed when the quantity of interest is small compared with the operators, as Table 7 shows: the entropy of an almost pure state is wrong by more than 100 % in single precision, and the round-off floor of a trace inequality at equality is several orders of magnitude higher.

Cost. An entry takes 4, 8, 8 or 16 bytes (`alg.bytes_per_operator(B)` reports the memory of a batch). A complex multiply–add costs about four real ones; for eigensolvers the measured ratio of complex to real is smaller, while double against single precision costs a factor that depends on the device (Table 6, measured on the GPU of the full run). Data-centre GPUs execute FP64 at half the FP32 rate, consumer GPUs at 1/32–1/64; the full-size runs of this paper used a consumer GPU. FP32 products may run in the TF32 format on recent GPUs; PyTorch keeps true FP32 by default, the library does not change this, and the paper scripts record the setting. We recommend the smallest type that is exact for the model and the precision that the size of the effect requires.

6.7 Safeguards

Batched computations make it easy to request hours of work or more memory than the device has. The `cost` module therefore times the first few items of every large batched decomposition and matrix exponential and warns with the expected total time when it exceeds a threshold (10 s by default). It compares the estimated peak memory of large allocations with the free device memory, warning above 50 % and raising `InsufficientMemoryError` above 90 %. After the second step of a time-stepping

type	B/entry	MB/batch	generate	eigh f(A)	SVD |A|	channel	expm	opt. step
`float32`	4	16	1	1	1	1	1	1
`float64`	8	32	1.2	7.7	21	3.7	2.1	3.4
`complex64`	8	32	1.1	1.6	1.8	1.8	1.5	1.8
`complex128`	16	64	3.2	11	65	13	5.4	11

Table 6: Bytes per entry, memory of one batch [MB], and time relative to `float32` for $C = 16$ sectors of size $k = 32$ and a batch of 256 operators (full run on NVIDIA GeForce RTX 5090; float32 times per sample: generate: 918 μs, eigh f(A): 7.24 μs, SVD |A|: 11.4 μs, channel: 1.83 μs, expm: 4.96 μs, opt. step: 70.9 μs).

type	$\max \lVert UU^* - 1 \rVert$	$\lVert \sqrt{A}^2 - A \rVert / \lVert A \rVert$	rel. error $S(\rho)$, $\rho \approx$ pure	$\lvert z \rvert$ at equality
`float32`	6.6×10^{-7}	2.0×10^{-6}	2.0×10^{0}	5.3×10^{-4}
`float64`	1.3×10^{-15}	4.0×10^{-15}	6.6×10^{-9}	1.4×10^{-12}
`complex64`	8.3×10^{-7}	2.3×10^{-6}	2.3×10^{0}	2.0×10^{-4}
`complex128`	1.6×10^{-15}	4.5×10^{-15}	8.9×10^{-9}	1.8×10^{-12}

Table 7: Accuracy of the same computations in the four storage types: Haar unitaries ($n = 64$), SVD square root, entropy of $\rho = (1 - \varepsilon)|0\rangle\langle 0| + \varepsilon\, 1/n$ with $\varepsilon = 10^{-7}$, and the round-off floor of $z = \mathrm{Tr}(Y|X|Y) - \mathrm{Tr}\,|YXY|$ for commuting X, Y (exactly 0); full run on NVIDIA GeForce RTX 5090.

or optimisation loop it warns with the expected duration. All thresholds can be changed, and the checks can be switched off locally.

7 Usage by example

The repository contains short complete programs in `man/article/listings`; the test suite runs them, so they stay in sync with the code. Each runs on a CPU in about half a minute or less; the nine programs `ex*.py` use a GPU when one is available. Table 8 lists the nine introductory examples; programs of later sections are described where they are used.

Several examples check exact statements. In `ex3`, the Kraus operators $\sqrt{1 - gN}$, $\sqrt{g}\, a_1$, $\sqrt{g}\, a_2$ ($0 \le g \le \frac{1}{2}$) map sector N to N and $N - 1$, and the classical part is the transition matrix of a pure death process, $T = \left(\begin{smallmatrix} 1 & g & 0 \\ 0 & 1-g & 2g \\ 0 & 0 & 1-2g \end{smallmatrix}\right)$ (columns: from $N = 0, 1, 2$). A unital positive map cannot increase the Michelson contrast, while the non-unital loss channel increases it for most samples. In `ex7` the ground state moves through the sectors until the chain is fully polarised above the saturation field $h_s = 2J$; `ex9` is the starting point of Section 12.

8 Validation against exact results

Table 9 compares Monte Carlo estimates (double precision) with exact values. The rows cover fourth moments of Haar unitaries and orthogonal matrices, and for traceless involutions $A = B$ and Haar U the value $\mathbb{E}\,\tau\big((AUBU^*)^2\big) = -1/(n^2 - 1)$ of the Weingarten calculus [28]. They also cover Page's formula for $\langle S(\rho_A) \rangle$ [24, 25, 26] and the mean purity $(k + r)/(kr + 1)$ of random states of rank $r \le k$ [27, 29]. The last column is the deviation in standard errors; for a correct sampler these should look like standard normal deviates. For the circular ensembles the Wigner surmise only approximates the exact spacing distribution, so no z-score is given; Figure 4 shows the distributions, with L^1

program	what it shows	key API
`listings/ex1_random_operators.py`	10^4 positive operators in $M_2 \oplus M_3 \oplus M_4$ with contrast 0.6 and Haar-random eigenvectors in every sector; the three traces and spectral functions	`TypeIAlgebra`, `operator_from_eigenvalues`, `michelson_contrast`, `Tr_blunt`, `tau_vN`, `apply_function`
`listings/ex2_states_measurement.py`	Lüders conditioning on a projective measurement moves all probability into one sector; expectation values do not depend on the trace that represents the state	`states.random_density_matrix`, `condition_on`, `sector_probabilities`, `density`
`listings/ex3_channels.py`	particle loss of two fermionic modes as a map between the sectors $N = 0, 1, 2$; unital and non-unital channels and the contrast	`InterSectorChannel.from_blocks`, `transition_matrix`, `random_mixed_unitary_channel`
`listings/ex4_dynamics.py`	spontaneous emission of a two-level atom by RK4 and by the exponential of the generator, against $P_e(0)e^{-\gamma t}$ and $\lvert\rho_{eg}(0)\rvert e^{-\gamma t/2}$	`dynamics.lindblad_evolve`, `dynamics.lindblad_channel`
`listings/ex5_page_entropy.py`	mean entanglement entropy of random pure states on $\mathbb{C}^m \otimes \mathbb{C}^n$ against Page's formula [24]	`tensor_product`, `partial_trace`, `states.von_neumann_entropy`
`listings/ex6_inequality_search.py`	Haar sampling against batched gradient ascent for $\sup_U(\lvert\operatorname{Tr}(XUY)\rvert - \operatorname{Tr}(XY)) = \lVert YX\rVert_1 - \operatorname{Tr}(XY)$ (Section 9)	`random_unitary_operator`, `expm`, `trace_norm`
`listings/ex7_spin_chain.py`	Heisenberg ring of ten spins: the sector of the ground state in a uniform field, the mean level-spacing ratio of 100 disorder realisations in one batch, entanglement entropies	`SpinChain`, `random_field_heisenberg`, `level_spacing_ratio`, `entanglement_entropy`
`listings/ex8_large_and_open_chains.py`	Lanczos ground state of a sector of dimension 12870, Krylov evolution of a domain wall, momentum sectors, particle loss through jumps between sectors	`xxz_sparse`, `krylov.ground_state`, `krylov.evolve`, `xxz_momentum`
`listings/ex9_qaoa_xy_mixer.py`	XY mixer on a ring of six qubits with amplitude damping, dephasing and post-selection on the Hamming weight	`SpinChain.xxz`, `lowering`, `algebra.central`, `condition_on`

Table 8: The introductory programs in `man/article/listings` of the repository.

distances 0.022, 0.007 and 0.006 for $\beta = 1, 2, 4$ from the surmise and level repulsion $P(s) \sim s^\beta$. The spontaneous-emission example agrees with the analytic solution to 2.2×10^{-16} (exact exponential) and 2.1×10^{-8} (RK4). The three traces satisfy their defining identities on random operators to round-off (Table 10), and $\mathrm{Tr}_{\mathrm{norm}} = C\,\tau_{\mathrm{vN}}$ holds to 9.0×10^{-16}.

9 From sampling to optimisation

The NumPy study [9, 4] sampled three trace functionals:

$$\begin{aligned}
&\text{Exp. 1:} \quad z = \lvert\operatorname{Tr}(XUY)\rvert - \operatorname{Tr}(XY), && X, Y \geq 0,\ U \in \mathcal{M} \text{ unitary},\\
&\text{Exp. 2:} \quad z = \operatorname{Tr}\lvert XY\rvert - \operatorname{Tr}(\lvert X\rvert\,\lvert Y\rvert), && X = UX_0,\ Y = VY_0,\ X_0, Y_0 \geq 0,\ U, V \in \mathcal{M} \text{ unitary},\\
&\text{Exp. 3:} \quad z = \operatorname{Tr}(Y\lvert X\rvert Y) - \operatorname{Tr}\lvert YXY\rvert, && X = X^*,\ Y \geq 0,
\end{aligned}$$

with $\mathrm{Tr} = \mathrm{Tr}_{\mathrm{blunt}}$, operators normalised to $\lambda_{\max} = 1$, and the contrasts of the operators as parameters (of $\lvert X\rvert$ and $\lvert Y\rvert$ for Exp. 2 and 3). Two observations motivate a different approach.

Sampling misses the extremes. For fixed X, Y the supremum in Exp. 1 is known exactly, $\sup_U z = \lVert YX\rVert_1 - \operatorname{Tr}(XY) > 0$. The best of 1 000 000 Haar samples falls short of it for every k, misses the violation for all pairs from $k = 4$ on, and moves further away as k grows (Table 11, Figure 5). Batched gradient ascent (128 starts, 500 steps) finds the supremum to a relative error of at most 3×10^{-5}. This is concentration of measure: the unitaries close to the maximiser form a set whose Haar measure decays rapidly with the dimension. The same effect explains why the Monte

quantity	parameters	exact	Monte Carlo	std. error	z
$\mathbb{E}\lvert U_{11}\rvert^4$, $U(n)$	$n = 2$	0.333333	0.333604	4.7×10^{-4}	+0.6
$\mathbb{E}\lvert U_{11}\rvert^2$, $U(n)$	$n = 2$	0.5	0.500277	4.6×10^{-4}	+0.6
$\mathbb{E}\lvert U_{11}\rvert^4$, $U(n)$	$n = 4$	0.1	0.10028	2.2×10^{-4}	+1.3
$\mathbb{E}\lvert U_{11}\rvert^4$, $U(n)$	$n = 8$	0.0277778	0.0277966	7.5×10^{-5}	+0.3
$\mathbb{E}\lvert U_{11}\rvert^4$, $U(n)$	$n = 16$	0.00735294	0.00735803	2.3×10^{-5}	+0.2
$\mathbb{E}\lvert U_{11}\rvert^4$, $U(n)$	$n = 32$	0.00189394	0.00189625	6.2×10^{-6}	+0.4
$\mathbb{E}\lvert U_{11}\rvert^2$, $U(n)$	$n = 32$	0.03125	0.0312403	4.8×10^{-5}	-0.2
$\mathbb{E}\lvert U_{11}\rvert^4$, $O(n)$	$n = 2$	0.375	0.375106	5.8×10^{-4}	+0.2
$\mathbb{E}\lvert U_{11}\rvert^2$, $O(n)$	$n = 2$	0.5	0.500155	5.6×10^{-4}	+0.3
$\mathbb{E}\lvert U_{11}\rvert^4$, $O(n)$	$n = 4$	0.125	0.125085	3.1×10^{-4}	+0.3
$\mathbb{E}\lvert U_{11}\rvert^4$, $O(n)$	$n = 8$	0.0375	0.0376573	1.3×10^{-4}	+1.2
$\mathbb{E}\lvert U_{11}\rvert^4$, $O(n)$	$n = 16$	0.0104167	0.0104046	4.3×10^{-5}	-0.3
$\mathbb{E}\lvert U_{11}\rvert^4$, $O(n)$	$n = 32$	0.00275735	0.0027558	1.2×10^{-5}	-0.1
$\mathbb{E}\lvert U_{11}\rvert^2$, $O(n)$	$n = 32$	0.03125	0.0312731	6.7×10^{-5}	+0.3
$\mathbb{E}\,\tau((AUBU^*)^2)$	$n = 4$	-0.0666667	-0.0661548	6.3×10^{-4}	+0.8
$\mathbb{E}\,\tau((AUBU^*)^2)$	$n = 8$	-0.015873	-0.0159874	2.8×10^{-4}	-0.4
$\mathbb{E}\,\tau((AUBU^*)^2)$	$n = 16$	-0.00392157	-0.00414221	1.4×10^{-4}	-1.6
$\langle s^2 \rangle$, COE	$n = 64$	1.27324	1.28582	3.7×10^{-4}	–
$\langle s^2 \rangle$, CUE	$n = 64$	1.1781	1.18024	2.4×10^{-4}	–
$\langle s^2 \rangle$, CSE	$n = 64$	1.10447	1.10392	2.0×10^{-4}	–
$\langle S(\rho_A) \rangle$	$m = 2, n = 16$	0.646516	0.646516	1.6×10^{-4}	0.0
$\langle S(\rho_A) \rangle$	$m = 8, n = 16$	1.83367	1.83388	1.7×10^{-4}	+1.2
$\langle S(\rho_A) \rangle$	$m = 16, n = 16$	2.27487	2.27495	1.4×10^{-4}	+0.6
$\mathbb{E}\,\mathrm{Tr}\,\rho^2$	$k = 4, r = 2$	0.666667	0.666312	3.2×10^{-4}	-1.1
$\mathbb{E}\,\mathrm{Tr}\,\rho^2$	$k = 4, r = 4$	0.470588	0.470523	2.1×10^{-4}	-0.3
$\mathbb{E}\,\mathrm{Tr}\,\rho^2$	$k = 16, r = 4$	0.307692	0.307726	6.4×10^{-5}	+0.5
$\mathbb{E}\,\mathrm{Tr}\,\rho^2$	$k = 16, r = 16$	0.124514	0.124527	1.7×10^{-5}	+0.8

Table 9: Monte Carlo estimates versus exact values (for $\langle s^2 \rangle$ the Wigner surmise, an approximation, hence no z); samples: 400000 per Haar moment, 400000 per Weingarten row, 20000 matrices per circular ensemble, 50000 states per Page row, 100000 per purity row (full run on NVIDIA GeForce RTX 5090, PyTorch 2.14.1+cu130).

trace	error of tr(1)	$\max \lvert \mathrm{tr}(AB) - \mathrm{tr}(BA) \rvert$	$\max \lvert \mathrm{tr}(UAU^*) - \mathrm{tr}(A) \rvert$	$\min \mathrm{tr}(A^*A)/\lVert A \rVert_F^2$
$\mathrm{Tr}_{\mathrm{blunt}}$	0	4.0×10^{-15}	5.9×10^{-15}	1.000
$\mathrm{Tr}_{\mathrm{norm}}$	0	1.4×10^{-15}	1.9×10^{-15}	0.223
τ_{vN}	0	4.4×10^{-16}	6.1×10^{-16}	0.074

Table 10: Identities of the three traces on 2000 random non-Hermitian operators in $M_2 \oplus M_3 \oplus M_5$ (double precision): normalisation $(\sum_c k_c, C, 1)$, tracial property, unitary invariance, and faithfulness; the last column is bounded below by $\min_c w_c$; full run on NVIDIA GeForce RTX 5090.

Carlo samples of [4] showed no violations for $k = 16$: it is a property of the sampling, not of the inequality.

Constrained optimisation. The natural objects are envelopes such as $\overline{z}(\delta) = \sup\{z : \Delta(X) = \delta\}$ and $\underline{z}(\delta) = \inf\{z : \Delta(X) = \delta\}$, taken over all remaining variables. The module `optimize` parametrises the constraint set exactly. The spectrum is $\lambda_{\max} = 1$, $\lambda_{\min} = (1-\delta)/(1+\delta)$, and the other eigenvalues $\lambda_{\min} + (1 - \lambda_{\min})(1 + \sin\theta)/2$ in between; unlike a sigmoid, this reaches both ends at finite θ, where

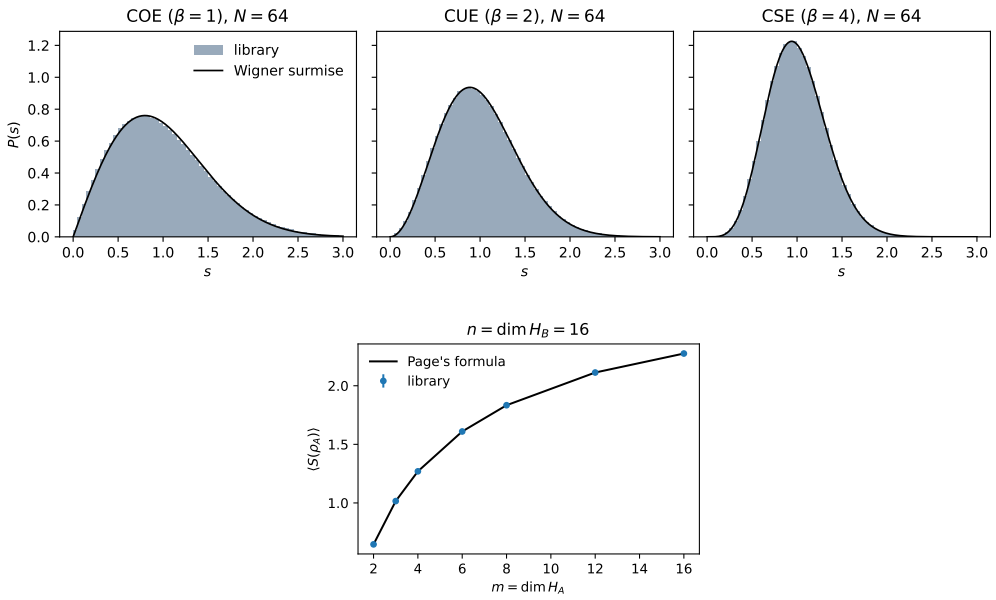


Figure 4: Top: nearest-neighbour spacings of COE, CUE and CSE eigenphases ($n = 64$, 20000 matrices per ensemble) against the Wigner surmise. Bottom: mean entanglement entropy of random pure states against Page's formula (50000 states per point) (error bars: two standard errors).

k	violation found		$\sup z/\operatorname{Tr} XY$	best sample	ascent rel. error		time [s]	
	sampling	ascent		$z/\operatorname{Tr} XY$	median	max	sampl.	ascent
2	14/16	16/16	7.1×10^{-4}	5.6×10^{-4}	1.2×10^{-8}	5.4×10^{-6}	64	2.8
4	0/16	16/16	2.3×10^{-3}	-0.12	4.4×10^{-8}	3.1×10^{-6}	66	2.8
8	0/16	16/16	5.6×10^{-3}	-0.48	2.9×10^{-7}	2.0×10^{-6}	70	2.8
16	0/16	16/16	6.7×10^{-3}	-0.72	2.4×10^{-6}	3.9×10^{-6}	88	2.7
32	0/16	16/16	8.9×10^{-3}	-0.86	7.4×10^{-6}	1.6×10^{-5}	1.8e+02	6.6
64	0/16	16/16	7.0×10^{-3}	-0.93	2.0×10^{-5}	2.8×10^{-5}	2.7e+02	11

Table 11: Exp. 1 for fixed random X, Y in $M_k(\mathbb{C})$ (eigenvalues uniform in $[0.2, 1.2]$, double precision): Haar sampling versus gradient ascent (medians over 16 pairs; full run on NVIDIA GeForce RTX 5090).

extremal spectra often lie. The positions of $\lambda_{\max}$ and $\lambda_{\min}$ are drawn per start, in the same sector or in different ones. The eigenvectors are $U = U_0 e^{G-G^*}$ with a Haar-random U_0 per start and a chart that is re-centred after every round. Discrete variables, such as the signs of the eigenvalues of a self-adjoint X (Exp. 3), are drawn per sector and improved by single sign flips between gradient rounds. Every iterate is admissible, and every grid point and start is one element of the batch, so each envelope is one batched optimisation.

Two implementation details matter. (a) In a real algebra a chart $U_0 e^{G-G^*}$ never leaves the connected component of U_0, and an extremiser with determinant $+1$ in every sector is reached from a Haar-random U_0 only with probability 2^{-C}. Half of the starts therefore begin at a structured point (diagonal X, Y rotated in pairs of eigenvectors of X, two-level spectra). With Haar-random starts only, a first run for $C = 32$ sectors of size 2 fell short of the true envelopes by up to 35%. With structured starts the searches of the companion paper [1] reproduce the known envelopes of the classical inequalities to 10^{-7}, and every closed form proved there to 3×10^{-4} or better. (b) Where the objective has a kink at the maximiser (a trace norm at a degenerate point), the remaining gap

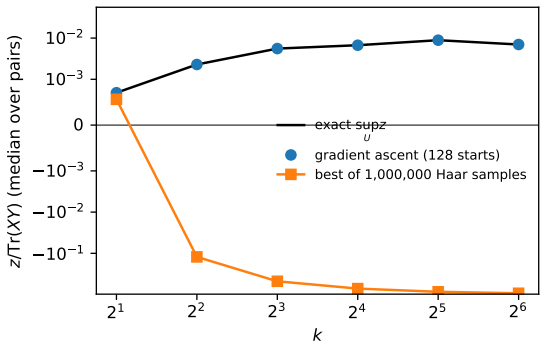


Figure 5: Best z found by sampling and by gradient ascent, relative to $\mathrm{Tr}(XY)$, against the exact supremum (symmetric log scale).

decreases with the step size. Applied to the three functionals, the optimiser located their envelopes for sectors of every size; the closed forms it suggested are proved in the companion paper [1].

Channels, generators and measurements as parameters. The same machinery applies to maps. In `optimize.StinespringParam` every input sector c carries an isometry into an environment of dimension r times the allowed output sectors. For the thin SVD $Z_c = U_c S_c W_c^*$ the code uses $V_c = U_c W_c^*$ directly, avoiding the squared condition number of $Z_c^* Z_c$; the Kraus operators are its blocks. For full column rank this equals $Z_c (Z_c^* Z_c)^{-1/2}$. At deficient rank the SVD completes an isometry within the allowed output space, so every value of the free matrices Z_c is a trace-preserving channel, within one algebra or between two algebras with a prescribed set of allowed sector transitions. Gradients require numerical full column rank and otherwise raise an error; at full rank the polar backward uses denominators $s_i + s_j$, which remain finite at repeated positive singular values. Gaussian Z_c give Haar-random isometries. `LindbladParam` does the same for GKSL generators (a self-adjoint Hamiltonian and free jump operators inside and between sectors), and `fit` minimises a loss over such parameters and leaves them at the fitted point. With the module `measurement` (POVMs, sampling of counts, likelihoods, instruments, Fisher information) this turns learning a channel or a generator from data into a call of the optimiser. In the test suite a channel of $M_1 \oplus M_3 \oplus M_2$ of Kraus rank two is recovered from twelve input–output pairs to 10^{-5}, and the generator of a damped and dephased qubit from two time points of six trajectories, predicting an unseen trajectory to 10^{-4}. These are small checks, not studies; tomography in symmetry algebras is the subject of ongoing work.

9.1 Rediscovering the data-processing thresholds of the Rényi divergences

A further test of the method is to ask it where the range of validity of a known inequality ends, without telling it. The module `divergences` evaluates the Petz divergences $D_\alpha(\rho\|\sigma) = \frac{1}{\alpha-1} \log \mathrm{Tr}(\rho^\alpha \sigma^{1-\alpha})$ [61] and the sandwiched divergences $\widetilde{D}_\alpha(\rho\|\sigma) = \frac{1}{\alpha-1} \log \mathrm{Tr}(\sigma^{\frac{1-\alpha}{2\alpha}} \rho\, \sigma^{\frac{1-\alpha}{2\alpha}})^\alpha$ [64, 65], batched and differentiable, with standard support conventions (implemented with relative numerical thresholds) and as logarithms of sums over sectors. Near $\alpha = 1$, within a precision- and log-norm-dependent window, the code evaluates $D + (\alpha - 1)V/2$, where D is the Umegaki relative entropy and V its relative entropy variance; the omitted term is $O((\alpha-1)^2)$. Single-precision inputs use double-precision work for scalar α near one and for tensor α, while retaining the input support tolerances and returning the input real dtype. It also provides the Petz recovery map [62, 63] and the universal recovery map of [70]. The data-processing inequality $D(\Phi(\rho)\|\Phi(\sigma)) \le D(\rho\|\sigma)$ holds for the Petz

	sandwiched $\widetilde{D}_\alpha$		Petz D_α	
α	dephasing	channels	dephasing	channels
0.2	3.8×10^{-1}	3.8×10^{-1}	–	–
0.25	4.3×10^{-1}	4.3×10^{-1}	–	–
0.3	4.4×10^{-1}	4.4×10^{-1}	–	–
0.35	4.0×10^{-1}	4.0×10^{-1}	–	–
0.4	3.1×10^{-1}	3.1×10^{-1}	–	–
0.45	1.7×10^{-1}	1.7×10^{-1}	–	–
0.48	7.0×10^{-2}	7.0×10^{-2}	–	–
0.49	3.5×10^{-2}	3.5×10^{-2}	–	–
0.5	–	–	–	–
0.51	–	–	–	–
0.52	–	–	–	–
0.55	–	–	–	–
0.6	–	–	–	–
0.7	–	–	–	–
0.8	–	–	–	–
0.9	–	–	–	–
1.1	–	–	–	–
1.25	–	–	–	–
1.5	–	–	–	–
1.75	–	–	–	–
1.9	–	–	–	–
1.95	–	–	–	–
1.98	–	–	–	–
2	–	–	–	–
2.02	–	–	–	1.7×10^{-1}
2.05	–	–	–	5.2×10^{-1}
2.1	–	–	–	9.9×10^{-1}
2.25	–	–	–	2.9×10^{0}
2.5	–	–	–	5.3×10^{0}
2.75	–	–	–	7.6×10^{0}
3	–	–	–	7.8×10^{0}

Table 12: Largest high-precision recomputed violation $v(\alpha)$ of the data-processing inequality on M_2 ("–": no violation above the tolerance 1×10^{-12} found; full run on CPU (4 threads)). The theorems guarantee $v \leq 0$ for $\alpha \geq \frac{1}{2}$ (sandwiched) and $\alpha \leq 2$ (Petz).

divergences for $\alpha \in (0,1) \cup (1,2]$ [61] and for the sandwiched ones for $\alpha \geq \frac{1}{2}$ [64, 65, 67, 66], and fails in general outside these ranges [64, 68, 69]. `scripts/renyi_threshold.py` maximises the violation $v(\alpha) = D(\Phi(\rho)\|\Phi(\sigma)) - D(\rho\|\sigma)$ on a grid of $\alpha \in [0.2, 3]$, over states on M_d and over the complete dephasing and all channels with a d-dimensional environment, with all values of α and all starts in one batch. Every reported violation is attained at an explicit point and recomputed in mpmath with 50 significant digits, with the channel re-normalised at that precision. A positive recomputed value is numerical evidence of a violation; without a guaranteed error enclosure it is not a rigorous numerical certificate. The absence of a violation is evidence only, since the search is local. The high-precision search finds a sandwiched violation at $\alpha = 0.499980$ with $v = 7.8 \times 10^{-5}$, and no violation above 1×10^{-12} was found at $\alpha = 0.500000$; the Petz inequality is violated at $\alpha = 2.000039$ with $v = 1.3 \times 10^{-4}$, and no violation was found at $\alpha = 2.000000$. The search brackets are $[0.499980, 0.500000]$ and $[2.000000, 2.000039]$, in agreement with $\frac{1}{2}$ and 2 ($d \in \{2, 3\}$, 64 starts and 600 steps per α; full run on CPU (4 threads)). Table 12 and Figure 6 show $v(\alpha)$ on the grid.

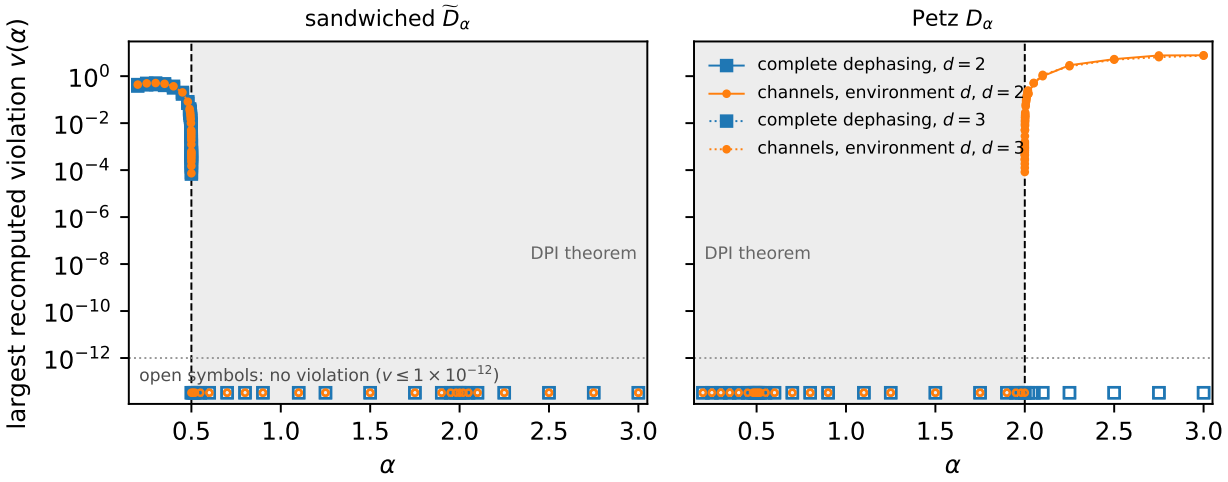


Figure 6: Largest high-precision recomputed violation $v(\alpha)$ for the sandwiched (left) and the Petz (right) divergences; dashed: the known thresholds; shaded: the range in which the data-processing inequality is a theorem.

10 Symmetric tensor products and spin chains

The basis of `SpinChain` is, by construction, that of the iterated fused product of single sites. Integrable chains provide exact results for every part of the module, and Table 13 lists those reproduced by `scripts/chains.py`; Table 16 and Figure 7 show some of them. Three entries need explanation. Peschel's formula gives the entanglement entropy of a block of the XX ground state from the eigenvalues ν of the restricted correlation matrix, $S = -\sum_\nu [\nu \ln \nu + (1-\nu)\ln(1-\nu)]$ [30], including the parity oscillations of open chains. Momentum sectors of periodic chains use the basis $|a,k\rangle = N_a^{-1/2} \sum_{j=0}^{L-1} e^{-ikj} T^j |a\rangle$ over orbit representatives a of the translation T (orbit length R_a, $N_a = L^2/R_a$; a contributes to k iff $kR_a \in 2\pi\mathbb{Z}$). The ground state of the antiferromagnetic Heisenberg ring has $k = 0$ for even and $k = \pi$ for odd $L/2$ (Marshall's sign rule). Blocks with $k \neq 0, \pi$ are complex in this basis even for real Hamiltonians, one of the cases where the field is fixed by the problem. With particle loss S_i^- on every site, any N-conserving Hamiltonian leaves the particle number a pure death process, so the sector distribution must be binomial.

For the Heisenberg chain with random fields $h_i \in [-W, W]$, the mean ratio of consecutive level spacings in the middle of the spectrum crosses over from the GOE value 0.5307 to the Poisson value $2\ln 2 - 1 \approx 0.3863$ as W grows [31, 32] (Figure 7, right). At these sizes this is the finite-size crossover used as a spectral diagnostic in studies of many-body localisation, and the batch dimension holds the 400 disorder realisations per point.

Beyond dense matrices. Lanczos with full re-orthogonalisation and restarts, batched over Hamiltonians, reproduces free-fermion and Heisenberg ground-state energies in sectors up to the sizes of Table 14. For a single Hamiltonian on a CPU, ARPACK [52] (`scipy.sparse.linalg.eigsh`) is the reference. Our solver re-orthogonalises fully for robustness and is batched over Hamiltonians, which is what lets disorder averages and GPUs pay off (Table 15).

computation	exact reference	size	max. deviation
open XX chain, all 2^L levels	free fermions, $\varepsilon_k = J\cos\frac{\pi k}{L+1}$	$L = 12$	7.1×10^{-15}
Heisenberg ring, levels of sector $N < L/2$	reappear in sector $N+1$ (SU(2) multiplets)		2.9×10^{-15}
domain-wall melting, XX chain (dense)	single-particle $\langle n_i(t)\rangle = [e^{iht}C(0)e^{-iht}]_{ii}$	$L = 16$, dim 12870	4.9×10^{-15}
domain-wall melting (Krylov)	same	$L = 22$, dim 705432	1.7×10^{-14}
block entropy of the XX ground state	Peschel's formula	$L = 14$	3.8×10^{-12}
momentum sectors, $N = L/2$	union of spectra = spectrum of the sector	$L = 14$	3.6×10^{-14}
particle loss S_i^-, any N-conserving H	binomial $p_n(t) = \binom{N_0}{n} e^{-n\gamma t}(1-e^{-\gamma t})^{N_0-n}$		2.4×10^{-10}
Heisenberg ring ground states	Bethe ansatz ($L = 4$), exact diagonalisation	Table 16	nine digits

Table 13: Exact results reproduced by the chain module (`scripts/chains.py`; full run on NVIDIA GeForce RTX 5090).

model	sector dim	E_0 (Lanczos)	E_0 (exact)	difference	time [s]
XX open, $L = 16$	12870	-4.9189757237	-4.9189757237	1.8×10^{-15}	0.14
XX open, $L = 20$	184756	-6.1907449998	-6.1907449998	8.9×10^{-16}	0.07
XX open, $L = 24$	2704156	-7.4629855550	-7.4629855550	2.7×10^{-15}	0.56
Heisenberg ring, $L = 16$	12870	-7.1422963606	-7.142296361	$< 5 \times 10^{-10}$	0.059

Table 14: Batched Lanczos on sparse sector Hamiltonians (time includes building the sparse Hamiltonian): ground-state energies against free fermions and the tabulated Heisenberg value; full run on NVIDIA GeForce RTX 5090.

11 Subalgebras, inclusions and symmetries

Three modules relate algebras to each other and to the symmetries that produce them: `inclusions` treats a subalgebra $\mathcal{N} \subset \mathcal{M}$ with its conditional expectations and indices, `decompose` computes the algebra generated by given matrices or by a finite group, and `su2` uses the full SU(2) symmetry of spin-$\frac{1}{2}$ chains. Table 17 lists the identities against which the test suite checks them.

Inclusions and conditional expectations. Let $\mathcal{N} = \bigoplus_{c=1}^{C} M_{k_c}$ and $\mathcal{M} = \bigoplus_{d=1}^{D} M_{m_d}$. Every unital inclusion $\mathcal{N} \to \mathcal{M}$ is, up to an inner automorphism $\operatorname{Ad} u$ of $\mathcal{M}$, of the form $\iota(x)_d = \bigoplus_c x_c \otimes 1_{\lambda_{dc}}$. It is given by the inclusion matrix $\Lambda = (\lambda_{dc}) \in \mathbb{Z}_{\geq 0}^{D\times C}$, $m_d = \sum_c \lambda_{dc} k_c$, whose entry λ_{dc} is the multiplicity of the summand c of $\mathcal{N}$ in the summand d of $\mathcal{M}$ [78]. `Inclusion` stores Λ and an optional batched unitary u; with the isometries $V_{dc,i}$ onto the copies of $\mathbb{C}^{k_c}$, ι is an `InterSectorChannel`. A faithful trace $\tau = \sum_d w_d \operatorname{Tr}_d$ on $\mathcal{M}$ restricts to $\mathcal{N}$ with weights $\nu = \Lambda^T w$, and

$$E(y)_c = \nu_c^{-1} \sum_{d,i} w_d \, V_{dc,i}^* u_d^* \, y_d \, u_d V_{dc,i}$$

is the unique τ-preserving conditional expectation onto $\mathcal{N}$, the adjoint of ι with respect to τ. For a connected inclusion the index is $[\mathcal{M} : \mathcal{N}] = \|\Lambda\|^2$ [77, 78], e.g. $(m/k)^2$ for $M_k \subset M_m$ and n for the diagonal subalgebra $\mathbb{C}^n \subset M_n$. The *Markov trace* is the trace whose weight vector is the Perron–Frobenius vector of $\Lambda\Lambda^T$; it is the trace for which the conditional expectation behaves as in a subfactor. The Watatani index is $\operatorname{Ind} E = \sum_j u_j u_j^*$ over a *quasi-basis*, a finite family $\{u_j\} \subset \mathcal{M}$

L	sector dim	Hamiltonians	`eigsh` [s]	library, GPU [s]	$\max\lvert\Delta E_0\rvert$
16	12870	1	0.0288	0.0552	6.2e-15
16	12870	20	0.621	0.114	4.3e-14
18	48620	1	0.115	0.0588	3.6e-15
20	184756	1	0.540	0.0642	1.1e-14

Table 15: Ground states of Heisenberg rings (one Hamiltonian) and random-field chains ($W = 3$, several Hamiltonians) in the sector $N = L/2$: ARPACK, one call per Hamiltonian, against the batched Lanczos solver; total time for all Hamiltonians (`scripts/baselines.py`; full run on NVIDIA GeForce RTX 5090).

L	dim ($N = L/2$)	E_0 (library)	E_0 (exact)	difference	E_0/L
4	6	-2.000000000	-2.000000000	$< 5 \times 10^{-10}$	-0.50000
6	20	-2.802775638	-2.802775638	$< 5 \times 10^{-10}$	-0.46713
8	70	-3.651093409	-3.651093409	$< 5 \times 10^{-10}$	-0.45639
10	252	-4.515446354	-4.515446354	$< 5 \times 10^{-10}$	-0.45154
12	924	-5.387390917	-5.387390917	$< 5 \times 10^{-10}$	-0.44895
14	3432	-6.263549534	-6.263549534	$< 5 \times 10^{-10}$	-0.44740
16	12870	-7.142296361	-7.142296361	$< 5 \times 10^{-10}$	-0.44639

Table 16: Ground-state energy of the Heisenberg ring $H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$; exact values to nine digits; E_0/L approaches the Bethe-ansatz value $\frac{1}{4} - \ln 2 \approx -0.4431$ (full run on NVIDIA GeForce RTX 5090).

with $\sum_j u_j E(u_j^* y) = y$ for all y [80]. It is central, $(\operatorname{Ind} E)_d = (\Lambda\Lambda^T w)_d / w_d$, and equals $[\mathcal{M} : \mathcal{N}]\,1$ exactly for the Markov trace. The Pimsner–Popa constant $\lambda(E) = \max\{\lambda : E(y) \geq \lambda y \text{ for } y \geq 0\}$ equals $\big(\max_d \sum_c (\nu_c/w_d)\min(k_c, \lambda_{dc})\big)^{-1}$ [79, Theorem 6.1]. For subfactors of II_1 factors it is $[\mathcal{M} : \mathcal{N}]^{-1}$, but for Type I inclusions it differs; the library also computes the complete version $\max\{\lambda : E - \lambda\,\mathrm{id} \text{ completely positive}\} = 1/\|\operatorname{Ind} E\|$ [82, 81]. For $\mathbb{C} \subset M_n$ the two are $1/n$ and $1/n^2$. Basic construction, Jones projection, Jones tower and relative commutant are also available, all batched and differentiable (`listings/inclusions.py`).

The spin-chain tower. The algebras $A_L = \bigoplus_{N=0}^{L} M_{\binom{L}{N}}$ of Section 10 form the tower $A_L \subset A_{L+1}$, $x \mapsto x \otimes 1_2$, whose Bratteli diagram is Pascal's triangle. Its inclusion matrix is the incidence matrix of the path with $2L + 3$ vertices, so

$$[A_{L+1} : A_L] = \|\Lambda\|^2 = 4\cos^2\frac{\pi}{2L+4} \;\nearrow\; 4,$$

the values $4\cos^2(\pi/n)$, $n = 4, 6, 8, \ldots$, of Jones' discrete series below 4 [77].

From a symmetry to the algebra: Wedderburn decomposition. Given matrices $S_1, \ldots, S_J \in M_n(\mathbb{C})$, or a unitary representation π of a finite group, `decompose.wedderburn` returns a unitary V with

$$V^* \mathcal{A} V = \bigoplus_c M_{k_c}(\mathbb{C}) \otimes 1_{m_c}, \qquad V^* \mathcal{A}' V = \bigoplus_c 1_{k_c} \otimes M_{m_c}(\mathbb{C})$$

for the von Neumann algebra $\mathcal{A}$ generated by the matrices, together with maps into the algebras of the library. The algorithm combines two randomised methods of Murota and coauthors [83, 85]. The eigenspaces of a random Hermitian element of the commutant $\mathcal{A}'$ are, generically, irreducible

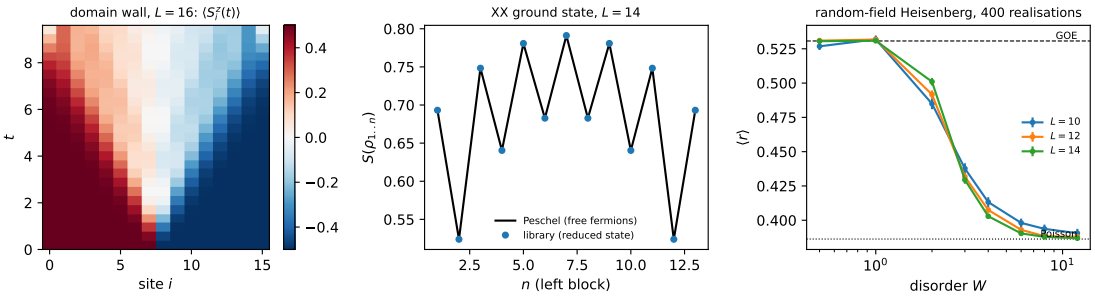


Figure 7: Left: $\langle S_i^z(t)\rangle$ after a domain-wall quench of the XX chain (light cone). Middle: entanglement entropy of the XX ground state, library vs. Peschel's formula. Right: mean level-spacing ratio of the random-field Heisenberg chain, with the GOE and Poisson values.

$\mathcal{A}$-submodules. For a second random $Y \in \mathcal{A}'$ and orthonormal bases Q_i, Q_j of two submodules, $Q_i^* Y Q_j$ intertwines them; by Schur's lemma it vanishes for inequivalent submodules and is a multiple of a unitary otherwise, which gives the equivalence classes and aligns their bases. Random elements of $\mathcal{A}'$ come from the group average $|G|^{-1}\sum_g \pi(g) X \pi(g)^*$ or from the null space of $X \mapsto ([H_j, X])_j$. Every result is verified before it is returned (generators in the decomposed algebra, fresh commutant elements in its commutant, and $\sum_c m_c^2 = \dim \mathcal{A}'$, which together prove equality by the double commutant theorem); a failed check triggers a new draw. For real generators, `real_structure` classifies every summand as real, complex or quaternionic from the sign of $J_c\overline{J_c}$, which has a gap of 2, and `real_basis` returns an adapted real basis [84]. `listings/symmetry.py` recovers Schur–Weyl duality [86] for four spins.

SU(2) symmetry of spin chains. The Heisenberg Hamiltonian $H = \sum J_{ij}\,\mathbf{S}_i \cdot \mathbf{S}_j$ commutes with all of SU(2). On $(\mathbb{C}^2)^{\otimes L} = \bigoplus_S \mathbb{C}^{m_S} \otimes V_S$ the commutant is $\bigoplus_S M_{m_S}(\mathbb{C})$, $m_S = \binom{L}{L/2-S} - \binom{L}{L/2-S-1}$, and an invariant Hamiltonian is $\bigoplus_S H_S \otimes 1_{2S+1}$: every multiplet is computed once, and the blocks shrink from $\binom{L}{L/2+S}$ to m_S. `SU2Chain` realises $\mathbb{C}^{m_S}$ as the highest-weight vectors in the $U(1)$ sector $N = L/2 + S$, in the *Yamanouchi basis*: built by adding sites one at a time with exact Clebsch–Gordan coefficients (Racah's formula [87]) and labelled by the sequence of intermediate total spins. It is orthonormal by construction and expressed in the $U(1)$ basis, where the block of an invariant operator is $A_S = B_S^* A_N B_S$; spin correlations follow from reduced matrix elements (Wigner–Eckart theorem [88]). For $L = 14$ the largest block shrinks from 3432 to 1001 and the cubic cost of diagonalising all blocks by a factor of about 48; the bases themselves ($1.0 \cdot 10^8$ numbers for $L = 16$) limit dense SU(2) computations to about $L = 16$.

11.1 Continuous centres: direct integrals

The algebras of the library have a finite centre $\mathbb{C}^C$, the L^∞ of a finite set with point masses. A finite type I algebra with separable predual is in general a direct integral $\int_X^\oplus M_{k(x)}\,d\mu(x)$ over a σ-finite measure space (X, μ): its centre is $L^\infty(X, \mu)$, its elements are bounded measurable fields $x \mapsto A(x)$, and $\int \operatorname{Tr} A(x)\,d\mu(x)$ is a trace on it (one for each measure in the class of μ). Examples are the translation-invariant observables of a lattice system, a direct integral over the Brillouin zone; averages over disorder, a direct integral over the realisations; and classical probability, with $k \equiv 1$. The class `DirectIntegral` discretises (X, μ) by nodes x_i with weights $w_i > 0$: product rules on boxes (Gauss–Legendre or midpoint, with a density $\mu = \rho\,dx$), the periodic rule on a torus with the

module	identities checked by the test suite
`chains`	S_i^z of `SpinChain(L)` equals $S_i^z \otimes 1$ formed with `kron` in the fused product of `SpinChain(L-1)` and a site
`inclusions`	E unital, completely positive, $E \circ \iota = \mathrm{id}$, bimodule property, adjoint of ι (three traces, Markov trace, random weights); Pimsner–Popa constants against brute force (bisection over rank-one y, Choi matrices); $e_{\mathcal{N}} y e_{\mathcal{N}} = E(y) e_{\mathcal{N}}$, $e_{\mathcal{N}} \in \mathcal{N}'$, Markov property of the basic construction; tower: path-graph spectrum and index for $L \le 8$, embedding $x \otimes 1_2$, conditional expectation = normalised partial trace
`decompose`	hidden block structures under random unitaries; Fourier basis of the regular representation of $\mathbb{Z}_n$; Schur–Weyl duality on $(\mathbb{C}^2)^{\otimes L}$, $L \le 6$ (same central projections, k_c and m_c exchanged); Kramers degeneracy in quaternionic sectors
`su2`	full spectra of open and periodic chains with field and next-nearest-neighbour couplings against dense diagonalisation ($L \le 12$, 10^{-10}); ring energies of Table 16 ($L \le 14$); Lieb–Mattis theorem [89]; the two degenerate singlets at the Majumdar–Ghosh point [90, 91]; Clebsch–Gordan coefficients against SymPy
`divergences`	dense references; independence of the trace convention; data processing inside the proven ranges; Petz's equality condition; the recovery bound of [70, Theorem 2.1]

Table 17: Identities checked by the test suite for the modules of Sections 9.1–11.

momenta n	error, $t = 0.5$	error, $t = 1$
8	9.1×10^{-5}	1.6×10^{-2}
16	1.2×10^{-7}	4.1×10^{-3}
32	6.4×10^{-13}	1.0×10^{-3}
64	$< 10^{-15}$	2.6×10^{-4}
128	$< 10^{-15}$	6.4×10^{-5}
256	$< 10^{-15}$	1.6×10^{-5}

Table 18: Ground-state energy per cell of the SSH chain, $H(k) = \left(\begin{smallmatrix} 0 & 1+te^{-ik} \\ 1+te^{ik} & 0 \end{smallmatrix}\right)$, from `DirectIntegral.torus(n)`, against $-\frac{2}{\pi}(1+t)E\left(\frac{4t}{(1+t)^2}\right)$ (complete elliptic integral E); `scripts/direct_integral_demo.py`.

normalised Haar measure, Monte Carlo samples, or the cells of a partition with their measures. It works in $\bigoplus_i M_{k(x_i)}$, one sector per node, and maps a field to $\bigoplus_i A(x_i)$, the trace to $\sum_i w_i \operatorname{Tr} A(x_i)$, a normal state $\int \operatorname{Tr}(\rho(x)\,\cdot\,)\,d\mu$ to a density matrix, the central projection of a measurable set to a sum of sector units, and an essential supremum to a maximum over the nodes. Products, the functional calculus, spectra, norms and channels that preserve the centre act fibrewise and are exact at the nodes; only integrals carry the error of the rule, and for fields that are constant on the cells of a partition they are exact. The momentum sectors of Section 10 are the case of a torus, with the n momenta of a periodic chain of n cells.

```
X = DirectIntegral.torus(64, dims=2)                    # [0, 2 pi) with dk/2pi, 64 momenta
H = X.field(bloch_hamiltonian, is_self_adjoint=True)    # one 2x2 sector per momentum
e0 = X.trace(H.apply_function(lambda w: w.clamp(max=0)))  # int Tr min(H(k), 0) dk/2pi
```

Listing 2: The ground-state energy per cell of the SSH chain as an integral over the Brillouin zone.

Listing 2 and Table 18 show the result for the SSH chain. For the gapped chain ($t = 0.5$) the integrand is analytic and periodic, and the error of the periodic rule falls exponentially, to rounding at $n = 64$. For the gapless chain ($t = 1$) the lower band touches zero at $k = \pi$, the integrand $-|2\cos(k/2)|$ has a kink, and the error falls as n^{-2} (fitted slope 2.00). With 10^4 Monte Carlo nodes the root-mean-square error over 20 seeds is 3.6×10^{-3}, the rate $n^{-1/2}$ of random sampling. All nodes go through one batched eigendecomposition of 2×2 blocks: 10^5 momenta take 0.10 s on a CPU with

k	continuum limit	c	fit	$h_{1,3}/2$	extrap.	v	extrap.	index
2	Ising	0.5000	0.5000	0.2500	0.2499	2.8284	2.8307	2.0000
3	tricritical Ising	0.7000	0.7000	0.3000	0.2997	2.9389	2.9306	2.6180
4	tetracritical Ising	0.8000	0.8003	0.3333	0.3308	3.0000	2.9824	3.0000

Table 19: Chains of $SU(2)_k$ anyons with $H = -\sum_i e_i$. Central charge fitted on periodic chains (sizes up to $L = 22$ for $k \le 3$ and 18 for $k = 4$) with the exact velocity; ratio of gaps and velocity of open chains (up to $L = 24$, resp. 20), extrapolated in $1/L$; Jones index of $TL_{L-1} \subset TL_L$, equal to $4\cos^2\frac{\pi}{k+2}$.

4 threads. The companion paper [1] uses the same picture to extend its theorem on direct sums to direct integrals: on every finite type I algebra, only the contrast of the density inside the fibres enters the violation envelopes.

11.2 Anyonic chains and Temperley–Lieb algebras

The algebras of anyon chains are finite direct sums of matrix algebras as well, with the total topological charge in place of the particle number. For L spin-$\frac{1}{2}$ anyons of $SU(2)_k$ the states are the fusion paths $a_0, a_1, \dots, a_L$ with $a_{i+1} = a_i \pm 1 \in \{0, \dots, k\}$ (twice the spin), paths on the Dynkin diagram A_{k+1}. The Temperley–Lieb generators [71]

$$(e_i)_{a',a} = \delta_{a_{i-1}a_{i+1}} \frac{\sqrt{S_a S_{a'}}}{S_{a_{i-1}}}, \qquad S_a = \sin\frac{(a+1)\pi}{k+2},$$

act on the charge $a = a_i$ between anyons i and $i+1$ and satisfy $e_i^2 = d e_i$, $e_i e_{i\pm1} e_i = e_i$ with the quantum dimension $d = 2\cos\frac{\pi}{k+2}$; e_i/d projects onto their trivial fusion channel. For an open chain from the charge 0 the last charge $b = a_L$ is the total charge, and the algebra generated by the e_i is $\bigoplus_b M_{n_b}$, n_b the number of paths from 0 to b: a `TypeIAlgebra` with one sector per total charge. `AnyonChain` orders the paths so that $TL_{L-1} \subset TL_L$ is an `Inclusion` in standard form, with Bratteli diagram A_{k+1}. From $L = k$ on, when all charges of both parities occur, its index is $\|\Lambda\|^2 = 4\cos^2\frac{\pi}{k+2}$, its Markov trace weights the sectors by the quantum dimensions, and its basic construction reproduces the block sizes of TL_{L+1}: Jones' subfactors [77, 78] appear as towers of the library's algebras (`tests/test_anyons.py`). A periodic chain (even L, since A_{k+1} is bipartite) has two sectors, the parities of a_0, which translation by one site exchanges.

The Hamiltonian $H = -\sum_i e_i$ rewards neighbouring anyons for fusing into the trivial channel. It is the critical RSOS chain of Andrews, Baxter and Forrester and Pasquier [74, 75], whose continuum limit is the unitary minimal model with central charge $c = 1 - \frac{6}{(k+1)(k+2)}$ [72]; for $k = 3$ it is equivalent to the golden chain of Fibonacci anyons of Feiguin et al. [76], with $c = \frac{7}{10}$. Table 19 recovers these data with the library alone, to within 1%: the central charge from periodic chains, $E_0(L) = e_\infty L - \pi v c/(6L)$ plus higher-order corrections (of order L^{-3} and from irrelevant operators), with the velocity $v = \pi\sin\gamma/\gamma$, $\gamma = \pi/(k+2)$; the velocity from the first excitation of open chains in the sector $b = 0$, which lies at level 2 of the vacuum tower; and the ratio of the lowest level of the sector $b = 2$ to that excitation, which tends to the Kac value $h_{1,3}/2$ of the boundary condition $(0, b)$, whose spectrum is the character $\chi_{1,b+1}$ [73]. For $k = 3$ the periodic sizes reach $L = 22$ (dimension 39603 per sector, restarted Lanczos of the module `krylov`) and give $c = 0.7000$ and $e_\infty = -1.2360680$, against $1-\sqrt{5} = -1.2360680$. The whole table takes 8 s on a CPU with 4 threads (`scripts/anyon_chain.py`).

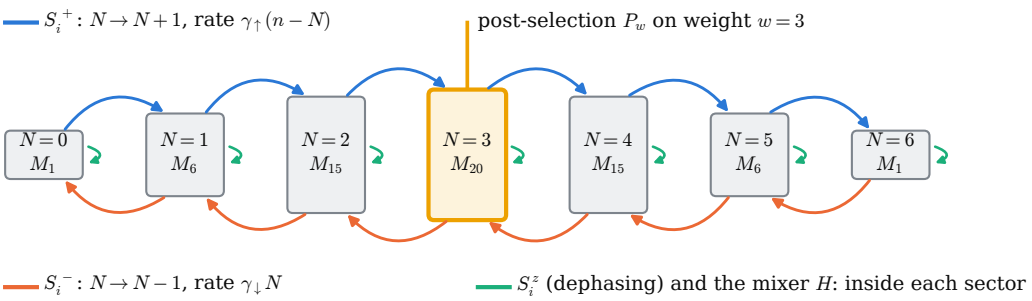


Figure 8: The sector ladder of a noisy register of $n = 6$ qubits. Sector N (Hamming weight) is the block $M_{\binom{n}{N}}$. Excitation S_i^+ moves up, amplitude damping S_i^- down, with total rates $\gamma_\uparrow(n-N)$ and $\gamma_\downarrow N$ that depend only on N. Dephasing and the mixer act inside the sectors, and post-selection projects onto the initial weight w.

12 Use case: noisy circuits with a conserved charge

Context. Quantum algorithms that preserve a charge offer a cheap test for errors. If the ideal evolution keeps the charge, a measurement of the charge at the end detects every error that changed it, and discarding those runs improves the accepted results. This *symmetry verification* [44, 45] is one of the standard error-mitigation techniques reviewed in [131]. Post-selection on the particle number was used, for instance, in Google's Hartree–Fock experiment [132]. Botelho et al. [130] proposed mid-circuit post-selection on Hamming-weight subspaces for QAOA. The XY mixers of the quantum alternating operator ansatz keep the Hamming weight [46, 128]; Streif et al. [129] derived the probability that such circuits stay in the feasible subspace under local depolarising noise and proposed codes that correct symmetry-breaking errors. In the language of open systems, the noisy register has a weak $U(1)$ symmetry in the sense of Buča and Prosen [133] and Albert and Jiang [134]. The Lindbladian commutes with conjugation by $e^{i\phi N}$, which says exactly that block-diagonal states stay block diagonal, i.e. that the dynamics never leaves $\mathcal{M} = \bigoplus_{N=0}^{n} M_{\binom{n}{N}}$. Without jumps that change N the symmetry is strong and every sector is conserved separately. In this language the measurement of the weight is a central projection P_w and post-selection is the Lüders conditioning on it, so the whole question is a computation in the algebra (Figure 8).

Model. A ring of n qubits evolves under the XY mixer $H = \frac{1}{4}\sum_{\langle ij\rangle}(X_iX_j+Y_iY_j) = \frac{1}{2}\sum_{\langle ij\rangle}(S_i^+S_j^- + S_i^-S_j^+)$ for a time $T = 2$ from $|1^w0^{n-w}\rangle$, $w = n/2$. Three kinds of local noise act on every site: amplitude damping S_i^- (rate $\gamma_\downarrow$), thermal excitation S_i^+ ($\gamma_\uparrow$) and dephasing S_i^z (γ_φ). With ψ the noiseless state at time T we compute the acceptance probability $p = \mathrm{Tr}(P_w\rho(T))$, the fidelity $F = \langle\psi|\rho(T)|\psi\rangle$ and the fidelity after post-selection $F_{\mathrm{post}} = F/p$. Let $\Gamma_w = \gamma_\downarrow w + \gamma_\uparrow(n-w)$ be the total rate of leaving sector w and $I = \int_0^T \sum_i \big(\frac{1}{4} - \langle S_i^z\rangle^2_{\psi(t)}\big)\,dt$ the dephasing integral along the noiseless trajectory.

Proposition 1. *Let H (possibly time dependent) commute with N, and let the register start in a pure state of weight w.*

(a) The weight distribution $p_N(t) = \mathrm{Tr}(P_N\rho(t))$ is that of the classical birth–death process with death rate $\gamma_\downarrow N$ and birth rate $\gamma_\uparrow(n-N)$; it does not depend on H or γ_φ.

(b) *If* $\gamma_\uparrow = 0$*:* $P_w \rho(T) P_w = e^{-\gamma_\downarrow w T} \rho_\varphi(T)$*, with* ρ_φ *the solution for* $\gamma_\downarrow = 0$*. In particular* $p = e^{-\gamma_\downarrow w T}$*, and* $F_{\mathrm{post}} = 1$ *if also* $\gamma_\varphi = 0$*.*

(c) *If* $\gamma_\varphi = 0$*:* $0 \le 1 - F_{\mathrm{post}} \le 1 - e^{-\Gamma_w T}/p = \gamma_\downarrow \gamma_\uparrow T^2 \big[w(n-w) + \frac{n}{2}\big] + O\big((n\gamma T)^3\big)$*,* $\gamma = \max(\gamma_\downarrow, \gamma_\uparrow)$*.*

(d) *If* $\gamma_\downarrow = \gamma_\uparrow = 0$*:* $p = 1$ *and* $1 - F_{\mathrm{post}} = 1 - F = \gamma_\varphi I + O(\gamma_\varphi^2)$*.*

(e) *To first order in all rates,* $1 - F = \Gamma_w T + \gamma_\varphi I$ *and* $1 - F_{\mathrm{post}} = \gamma_\varphi I$*; for* $\gamma_\varphi > 0$ *the gain* $(1 - F)/(1 - F_{\mathrm{post}})$ *tends to* $1 + \Gamma_w T/(\gamma_\varphi I)$ *as the rates go to zero at fixed ratios.*

Proof. Since $S_i^\pm P_N = P_{N\pm1} S_i^\pm$ and $\sum_i S_i^+ S_i^- = N$, $\sum_i S_i^- S_i^+ = n - N$ are central, $\dot p_N = \gamma_\downarrow[(N+1)p_{N+1} - N p_N] + \gamma_\uparrow[(n-N+1)p_{N-1} - (n-N)p_N]$; H and the dephasing jumps commute with P_N and do not contribute, which is (a). Expand $\rho(T)$ in the jumps $S_i^\pm$ (Dyson series). The no-jump generator differs from the Lindbladian with $\gamma_\downarrow = \gamma_\uparrow = 0$ by $-\frac{1}{2}\{\gamma_\downarrow N + \gamma_\uparrow(n-N), \cdot\}$, which on sector w is the scalar $-\Gamma_w$. Hence the part of $\rho(T)$ in sector w is $e^{-\Gamma_w T}\rho_\varphi(T)$ plus terms with $m \ge 1$ up and m down jumps. (b) Without up jumps only the first term remains. (c) For $\gamma_\varphi = 0$, $\rho_\varphi(T) = |\psi\rangle\langle\psi|$, and the remaining terms form $J \ge 0$ with $\operatorname{Tr} J = p - e^{-\Gamma_w T}$. Then $F_{\mathrm{post}} = (e^{-\Gamma_w T} + \langle\psi|J|\psi\rangle)/p \ge e^{-\Gamma_w T}/p$. The leading terms of $\operatorname{Tr} J$ are one down and one up jump in either order, with probabilities $\gamma_\downarrow w\, \gamma_\uparrow (n-w+1) T^2/2$ and $\gamma_\uparrow (n-w)\, \gamma_\downarrow (w+1) T^2/2$. (d) Dephasing keeps the sector, and the first-order term gives $1 - F = \gamma_\varphi \int_0^T \sum_i (\langle (S_i^z)^2\rangle - \langle S_i^z\rangle^2)\, dt$ with $(S_i^z)^2 = \frac{1}{4}$. (e) Every first-order jump $S_i^\pm$ leaves sector w and contributes 0 to F, so $F = e^{-\Gamma_w T}(1 - \gamma_\varphi I) + O(\gamma^2)$; with post-selection only the dephasing term remains at first order, by (c) and (d). □

None of these statements is new in substance. Statement (b) is the standard reason why particle-number post-selection removes pure loss: N is central, so damping commutes with the post-selection. Statements (c)–(e) make explicit what remains. Thermal noise is suppressed from first to second order with a coefficient $w(n-w) + \frac{n}{2} = n(n+2)/4$ at half filling, and dephasing is not detected at all. We state the proposition because it fixes what an exact computation must reproduce and isolates the two numbers it cannot predict: the integral I, and the fraction of the bound in (c) that is actually lost.

Computation. Listing 3 shows the core of the computation for zero temperature. All noise models and rates form one batch: 28 settings per n (four noise models, seven rates between 10^{-3} and 10^{-1}), with the rates folded into the coefficients of the jumps. One call of `lindblad_evolve` with the completely positive splitting (`method='kraus'`, 40 steps) evolves the batch. The jumps are `SectorJumps` (Section 6.4); as dense maps between sectors they would need $3n$ tensors of $(n+1)^2 \binom{n}{n/2}^2$ entries, 2.3 GB each for $n = 12$. Table 20 and Figure 9 show the results for $n = 4, \ldots, 10$ (script `scripts/noisy_xy_mixer.py`).

Results. The computation confirms the proposition in every case. At zero temperature the post-selected state agrees with the noiseless one to 3×10^{-15}, and the acceptance agrees with $e^{-w\gamma T}$ to 5×10^{-5}, the error of the time steps. For all noise models the acceptance agrees with the birth–death process of (a) to 3×10^{-4}. The thermal infidelity after post-selection has slope 2.00. For $n = 4, 6, 8, 10$ it is 0.73, 0.76, 0.81 and 0.84 times the bound of (c), independently of the rate. The complementary fraction is the overlap with ψ that a state retains after one up and one down jump. The dephasing infidelity agrees with $\gamma_\varphi I$ to 0.1%. The integral I saturates with n at fixed T (1.17 for $n = 4$, 1.43 for $n = 10$): by time T only the qubits near the two domain walls of $|1^w 0^{n-w}\rangle$ have moved, and dephasing hurts only where the state is not a product of S^z eigenstates. The detected error grows with the weight w and the undetected one does not, so post-selection gains more for larger registers.

```
import torch
from torch_vn_algebra import SpinChain, dynamics, states

n, w, T = 6, 3, 2.0
q = SpinChain(n, 'periodic', precision='double')   # sectors: Hamming weight N = 0..n
H = q.xxz(J=1.0, Delta=0.0)                         # XY mixer
rho0 = q.basis_state('111000')
psi = dynamics.von_neumann(rho0, H, [T])[0]         # noiseless reference state
g = torch.logspace(-3, -1, 7, dtype=torch.float64)  # seven damping rates: one batch
jumps = [q.sector_jump(i, 'lower').scaled(g.sqrt()) for i in range(n)]   # S^-_i
rho = dynamics.lindblad_evolve(rho0, H, jumps, [0.0, T], substeps=40, method='kraus')[-1]
P = q.algebra.central([float(N == w) for N in range(n + 1)])  # weight-w projection
post, p = rho.condition_on(P)                       # post-selection (Lueders rule)
F_post = states.trace_of_product(post, psi).real
print(p / torch.exp(-w * g * T) - 1, 1 - F_post)   # both ~ 0
```

Listing 3: Amplitude damping of a six-qubit XY mixer for seven rates in one batch, and post-selection on the Hamming weight (Proposition 1(b)): both printed deviations are at the level of the time-step error and of round-off.

					(b) T_1, zero temp.		(c) thermal	(d) dephasing		mixed, $\gamma \approx 0.022$	
n	$\binom{n}{w}$	$\sum_N \binom{n}{N}^2$	4^n	time [s]	err. p	$\lvert 1-F_{\mathrm{post}}\rvert$	slope	I	$\frac{1-F_{\mathrm{post}}}{\gamma_\varphi I}$	$1-F$	$1-F_{\mathrm{post}}$
4	6	70	256	0	4e-7	9e-16	2.00	1.167	0.999	0.1126	0.0254
6	20	924	4096	1	6e-6	1e-15	2.00	1.380	0.999	0.1573	0.0307
8	70	12870	65536	18	2e-5	9e-16	2.00	1.421	0.999	0.1969	0.0328
10	252	184756	1048576	930	5e-5	3e-15	2.00	1.426	0.999	0.2340	0.0345

Table 20: Noisy XY mixer on a ring of n qubits, $T = 2$ (CPU, double precision). Dimensions: the weight sector, the algebra and a dense density matrix; time for all 28 settings. (b) Relative deviation of p from $e^{-w\gamma T}$ (the error of the second-order splitting) and $|1 - F_{\mathrm{post}}|$, maxima over the rates. (c) Slope of $\log(1 - F_{\mathrm{post}})$ against $\log\gamma$ for $\gamma_\uparrow = 0.2\gamma_\downarrow = 0.2\gamma$, at small γ. (d) The integral I and the ratio of the computed infidelity to its first-order value at $\gamma_\varphi = 10^{-3}$. Mixed: $\gamma_\downarrow = \gamma_\varphi = \gamma$, $\gamma_\uparrow = 0.1\gamma$ at $\gamma = 10^{-5/3} \approx 0.022$, without and with post-selection. Full sizes of the study, CPU (4 threads).

At $\gamma = 10^{-3}$ the gain for the mixed model is 4.75 for $n = 4$ and 8.61 for $n = 10$, within 1.3 % of the first-order value of (e). At $\gamma \approx 0.022$ it is 4.4 and 6.8 (Table 20). For algorithm design this quantifies the value of the symmetry: it protects against energy relaxation, not against dephasing, and the remaining error is predicted to first order by one integral over the noiseless trajectory.

What the computation adds, and what it does not cover. The computation gives exact values, up to the time-step error, for all noise models and rates in one batched run per register. It also gives the integral I and its saturation in n, the fraction of the thermal bound that is realised, and a gain that grows with n; none of these follows from the proposition alone. It does not model a QAOA circuit as practised. There is no phase separator, no layering, a single evolution time and a product initial state rather than a Dicke state. Phase separators are diagonal and commute with N, so the proposition still holds for layered circuits, but I and the gain depend on the circuit and would have to be recomputed. Two-qubit gate noise, readout errors and coherent errors are not modelled. Single-qubit depolarising noise does stay in the algebra: its Pauli-X and Y parts combine into $S^{\pm}$ jumps of equal rate, $\mathcal{D}[X] + \mathcal{D}[Y] = 2(\mathcal{D}[S^+] + \mathcal{D}[S^-])$, so it is the thermal model at $\gamma_\uparrow = \gamma_\downarrow$ plus dephasing. Noise that is not $U(1)$-covariant leaves the algebra: bit flips X_i alone, unequal X and Y rates, or coherent X rotations. For instance $X\rho X$ contains $S^+\rho S^+$, which maps the block (w, w)

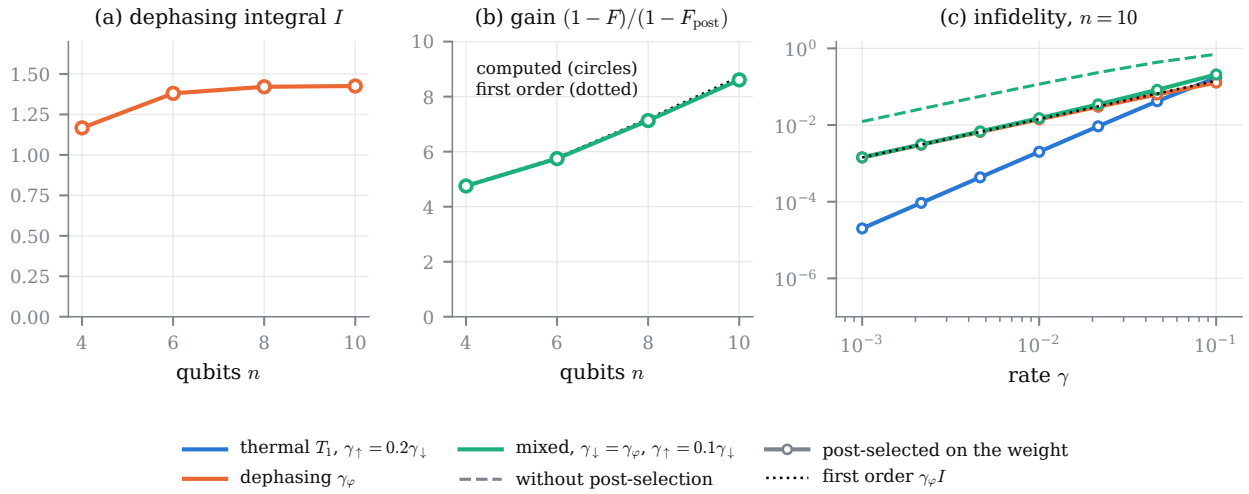


Figure 9: Noisy XY mixer, $T = 2$. (a) The dephasing integral I saturates with n. (b) The gain of post-selection for the mixed model at $\gamma = 10^{-3}$ grows with n and follows the first-order value $1+\Gamma_w T/(\gamma_\varphi I)$ of Proposition 1(e). (c) Infidelities for $n = 10$: post-selection suppresses thermal noise to second order and leaves dephasing, which follows $\gamma_\varphi I$; for zero-temperature damping (not shown) the post-selected state is exact.

to the off-diagonal block $(w+1, w-1)$; such noise cannot be represented in (1) and needs the full M_{2^n}, which is a limitation of the approach. Finally, we make no claim of speed for this example. The run for $n = 10$ took 930 s on a CPU for 28 settings, dominated by dense products of the padded Hamiltonian (Section 16). A single density matrix of dimension 2^{10} is within easy reach of dedicated solvers [19, 94, 21]; comparisons with them are left for future work.

12.1 Noise-aware state preparation with post-selection

Section 12 evaluated a fixed evolution. In practice the evolution is designed, and the design can take into account the noise and the fact that the Hamming weight will be measured. XY-QAOA needs an initial state in the feasible subspace, and the Dicke state $|D_w^n\rangle$, the uniform superposition of all strings of weight w, is the standard choice [46, 128]. Streif et al. [129] computed how likely such circuits are to stay in the feasible subspace under local noise. Botelho et al. [130] proposed post-selection on Hamming-weight subspaces for QAOA, an instance of symmetry verification [131]. This raises a design question. Does a preparation that will be post-selected need different controls from one that will not, and how much does designing for the noise gain over a noiseless design? The question is an optimisation over the controls of a Lindblad evolution in $\mathcal{M} = \bigoplus_N M_{\binom{n}{N}}$, and its objective uses the centre of $\mathcal{M}$. It needs gradients through open-system dynamics in the sectors, the central projection P_w, and many noise settings at once. These are the three things the library provides.

Parametrisation. A ring of n qubits starts in $|1^w 0^{n-w}\rangle$, $w = n/2$, and is driven during a time T by $S = 4$ piecewise-constant segments of length T/S with Hamiltonians

$$H_s = \sum_{\langle ij\rangle} J_{ij,s}\, \tfrac{1}{2}\big(S_i^+ S_j^- + S_i^- S_j^+\big) + \sum_i h_{i,s} S_i^z + V_s \sum_{\langle ij\rangle} S_i^z S_j^z, \qquad |J_{ij,s}|, |h_{i,s}|, |V_s| \le 1, \quad 0 < T \le 2n. \tag{3}$$

n	design	T	F_0	$\bar{p}$	$\bar{F}$	$\bar{F}_{\mathrm{post}}$	$\min_k F_k$	$\min_k F_{\mathrm{post},k}$
4	(i)	2.73	1.0000	0.941	0.929	0.987	0.795	0.948
4	(ii)	2.66	0.9993	0.942	0.930	0.988	0.801	0.951
4	(iii)	2.71	1.0000	0.941	0.930	0.988	0.799	0.952
6	(i)	5.49	0.9981	0.846	0.807	0.952	0.490	0.816
6	(ii)	4.88	0.9889	0.860	0.821	0.953	0.535	0.846
6	(iii)	5.14	0.9949	0.854	0.818	0.956	0.519	0.840

Table 21: Dicke-state preparation on a ring of n qubits by the three designs, evaluated over the 12 noise settings (four models, $\gamma \in \{0.003, 0.01, 0.03\}$). F_0: fidelity without noise. Means and minima over the settings; every value is from an independent evaluation with 20 steps per segment. Full run, CPU (4 threads).

These are bond-resolved XY couplings, local fields and a uniform phase-separator term. Every term commutes with N, so the controls stay in $\mathcal{M}$. The bond terms are `SpinChain.bond(i, j, Jxy, Jz)`, and H_s is a linear combination of them with trainable coefficients. Without the fields the target cannot be reached. The global spin flip $\prod_i X_i$ commutes with all bond terms and maps sector $n/2$ to itself, the Dicke state is even under it, and $|1^w 0^{n-w}\rangle$ has equal weight on both parities, so $F \le \frac{1}{2}$. With $V_s = 0$ as well, the evolution is quadratic in Jordan–Wigner fermions and cannot leave the Slater determinants. The noise consists of the four models of Section 12 (T_1 at zero temperature, thermal, dephasing and mixed), each at the rates $\gamma = 0.003$, 0.01 and 0.03, so the batch has $K = 12$ settings. Each segment is integrated over unit time with H_s replaced by $(T/S)H_s$ and jump coefficients $\sqrt{\gamma T/S}$ (`SectorJump.scaled`), so $\rho(T)$ is differentiable in all controls and in T.

Objectives. With $F_k = \langle D_w^n | \rho_k(T) | D_w^n \rangle$ for noise setting k, $p_k = \mathrm{Tr}(P_w \rho_k(T))$ the acceptance and $F_{\mathrm{post},k} = F_k/p_k$ the fidelity of the Lüders-conditioned state (`condition_on`), we compare three designs: (i) the noiseless design, which maximises $F_0 - \lambda T/T_{\max}$ without noise, $\lambda = 0.05$. The penalty pushes T close to the shortest time at which the controls reach $F_0 \approx 1$, which is the most favourable choice for a design that ignores the noise; (ii) the noise-aware design without post-selection, which maximises $\bar{F} = \frac{1}{K}\sum_k F_k$; and (iii) the noise-aware design with post-selection, which maximises $\bar{F}_{\mathrm{post}} = \frac{1}{K}\sum_k F_{\mathrm{post},k}$. By Proposition 1(a), p_k depends on T and the rates but not on the controls. At fixed T, therefore, $F_k = p_k(T) F_{\mathrm{post},k}$, and (ii) and (iii) differ only in the weights p_k of the settings and in the choice of T: design (ii) pays for the detected loss $\Gamma_w T$, design (iii) does not. Each objective is maximised by `optimize.extremize` (Adam, 150 steps; 400 for (i)). Every evaluation is a single batched `lindblad_evolve` per segment for all starts and all K settings (`method='kraus'`, 2 steps per segment). The noiseless design is the best of 32 random starts. Designs (ii) and (iii) start from the 8 ($n = 4$) or 4 ($n = 6$) best noiseless solutions, and the starts of (iii) also include the solution of (ii).

Results. Table 21 and Figure 10 show the outcome. For $n = 6$ the three designs use the times $T = 5.49$, 4.88 and 5.14. Design (ii) is the fastest and gives up the most noiseless fidelity ($F_0 = 0.9889$ against 0.9981 for (i)), because every unit of time costs it the detected loss as well. Design (iii) ignores that loss, so it spends more time than (ii) but still less than (i), because the undetected part of the noise (dephasing, and pairs of an up and a down jump) also grows with T. The order of the times is the same for $n = 4$. Each noise-aware design wins on its own objective. Design (ii) has the largest mean raw fidelity, $\bar{F} = 0.821$ against 0.807 for (i), and design (iii) the largest mean post-selected fidelity, $\bar{F}_{\mathrm{post}} = 0.956$ against 0.952. As factors on the infidelity, the noise-aware gains are modest: 1.07 for (ii) and 1.09 for (iii) at $n = 6$, and 1.02 and 1.07 at $n = 4$. In the worst setting of the batch the post-selected fidelity rises from 0.816 for (i) to 0.840 for (iii). Figure 10(b) shows the price. The

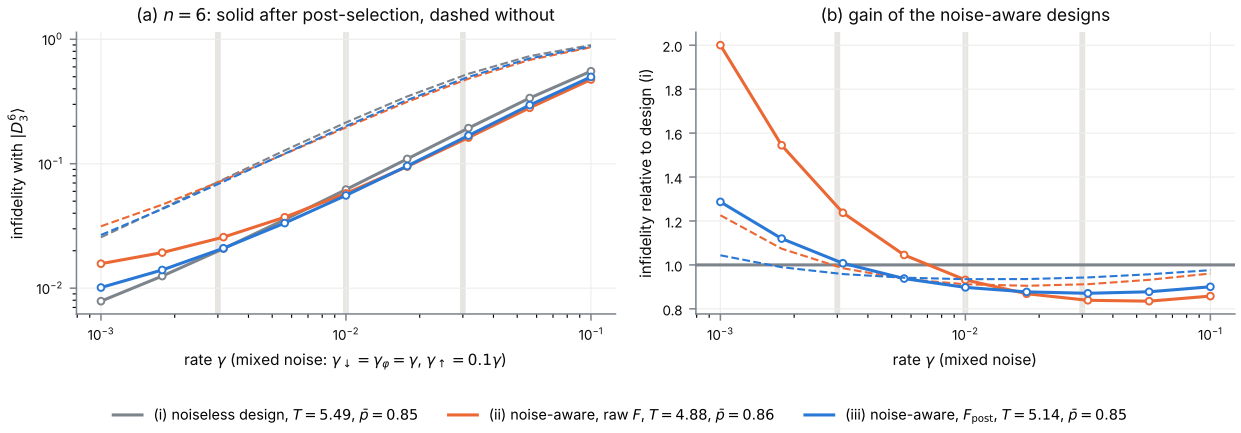


Figure 10: Designs (i)–(iii) for $n = 6$ under mixed noise ($\gamma_\downarrow = \gamma_\varphi = \gamma$, $\gamma_\uparrow = 0.1\gamma$) at rates inside and outside the training batch (grey bars). (a) Infidelity with $|D_w^n\rangle$ with (solid) and without (dashed) post-selection. (b) The same, relative to the noiseless design (i).

noise-aware designs are worse than (i) at rates below the training batch (factor 1.29 at $\gamma = 10^{-3}$ for (iii)) and better above it (0.90 at $\gamma = 0.1$). Post-selection itself is far stronger than any choice of design. For the noiseless design it lowers the mean infidelity by a factor 4.0 ($n = 6$) and 5.6 ($n = 4$), and from the best design without post-selection, (ii), to design (iii) with it the factor is 4.07 at $n = 6$. The price is the acceptance, $\bar{p} = 0.854$ for (iii) against 0.860 for (ii); it differs between the designs only through T. In short, once the noiseless design is as fast as the controls allow, little is left for noise-aware shaping. The centre does most of the work: the measurement of the weight removes the detected errors, and the optimiser then trades the remaining, undetected ones against time.

Checks. Every number in the table comes from a separate forward evaluation, not from the optimiser. It integrates in physical time, passes the rates as `rates` rather than as scaled coefficients, and uses 20 steps per segment. The optimiser's own values differ from it by at most 6×10^{-4}, the error of the coarse time steps. The acceptances agree with the birth–death process of Proposition 1(a) to 4×10^{-6}. At one point (design (iii), mixed noise, $\gamma = 0.03$, $n = 6$), `method='exact'` reproduces F, F_{post} and p to 6×10^{-6}. A directional derivative of the objective of (iii) agrees with central differences to a relative 6×10^{-8}. The whole study ($n = 4$ and 6) took 658 s on a CPU with 4 threads (script `scripts/noisy_xy_control.py`).

What this shows, and what it does not. The study shows that the library handles the whole loop of noise-aware design: a Lindblad evolution in symmetry sectors with sparse jumps, differentiated with respect to Hamiltonian controls and the duration; an objective defined through a central projection; and a batch of noise models, rates and starts in a single evolution. It also gives a quantitative answer for this model. With a time-optimal noiseless design as the reference, the noise-aware designs gain only a few per cent of the infidelity, while post-selection gains a factor of several. Proposition 1(a) explains why the two noise-aware objectives give such similar results: post-selection changes the objective only through T and the weights p_k. The comparison depends on the reference. With a weaker time penalty λ, the noiseless design stays well above the shortest time, and the same optimisation then reports larger noise-aware gains, which are really gains in speed. The search is local: there are few starts (4 for the noise-aware designs at $n = 6$), S is small, the controls are bounded and piecewise constant, and the reported designs are good solutions, not proven optima. The sizes are small ($n = 4, 6$). For $n = 8$

layer	map	network analogue
`ChannelLayer`	Stinespring isometry per input sector, within one algebra or between algebras with allowed sector transitions	linear layer plus contraction; width = Kraus rank; sparsity = sectors
`UnitaryLayer`	$\rho \mapsto U\rho U^*$, $U = e^{-iH}$, $H = H^* \in \mathcal{M}$	unitary layer [136]
`LindbladLayer`	$\rho \mapsto e^{t\mathcal{L}}\rho$, $\mathcal{L}$ a free GKSL generator	continuous depth [135]
`Residual`	$\rho \mapsto (1-p)\rho + p\,\Phi(\rho)$	skip connection [141]
`PostSelect`	$\rho \mapsto P\rho P/\operatorname{Tr}(P\rho P)$, $P \in \mathcal{M}$ a projection (central: a set of sectors)	nonlinear activation; cost $\operatorname{Tr}(P\rho P)$
`FeedForward`	instrument, outcome in the centre, branch per outcome	fan-out skip, classical only

Table 22: Layers of `torch_vn_algebra.learn`. Every layer is a `torch.nn.Module` that maps a batch of states to a batch of states; every value of its parameters is a physical map, with post-selection conditioned on acceptance above its numerical cutoff. `LindbladLayer` accepts the exact exponential or the CPTP Kraus splitting.

the padded 70×70 blocks of all nine sectors make every gradient step much more expensive, and we did not attempt that size on a CPU. The noise models are the idealised local ones of Section 12, with rates measured in units of the coupling bound. Gate-level circuits, readout errors and non-covariant noise are not modelled, and nothing here is a claim about particular hardware.

13 Quantum channels as trainable networks

The previous sections optimise a handful of parameters. The same differentiable objects can be stacked into networks whose layers are channels, and trained on data. This section describes the module `learn` that does this, the dictionary behind it, and two experiments that show what the algebra contributes: tomography in the algebra as training (Section 13.1), and residual networks of channels as Markovian evolutions (Section 13.2).

The dictionary. A channel is a linear map of density matrices, so a network of channels is a deep *linear* network with a constraint on every layer, and the analogy with neural networks is precise only where the constraint is respected. By Stinespring's theorem [139] a channel is an isometry into a larger space followed by a partial trace: a linear layer followed by a contraction, whose width is the Kraus rank. Composition of channels is depth; a Lindblad evolution $e^{t\mathcal{L}}$ is a block of continuous depth, the open-system form of a neural ODE [135, 140]; a conditional expectation onto a subalgebra is a pooling layer. The sector structure of (1) is a structured sparsity of the weights: a channel that respects the sectors has Kraus operators with blocks only on the allowed transitions, and correspondingly fewer parameters. Table 22 lists the layers.

Every parameter value is a channel. A `ChannelLayer` holds free matrices Z_c and uses the isometries from the direct thin-SVD polar factor (`StinespringParam` of the module `optimize`), with isometric completion at deficient rank as described in Section 9; a `LindbladLayer` holds a free Hamiltonian and free jump operators, i.e. a Cholesky-type factor of the Kossakowski matrix (`LindbladParam`). No constraint is imposed and no projection is needed, so any optimiser of `torch.optim` trains the network without leaving the set of channels. The channel gradient uses the polar backward and requires numerical full column rank. A `LindbladLayer` accepts only `exact` (the matrix exponential) or `kraus` (the CPTP splitting); it rejects RK4, whose steps need not preserve positivity. Gradients through the dynamics follow the chosen propagator; the Kraus splitting also uses the Daleckii–Krein

```
import torch
from torch import nn
from torch_vn_algebra import TypeIAlgebra, channels, learn, states

alg = TypeIAlgebra([1, 3, 2], [1, 3, 2], precision='double')
X = states.random_density_matrix(alg, batch_size=16)          # inputs
Y = channels.random_channel(alg, kraus_rank=3)(X)               # outputs of an unknown channel
net = nn.Sequential(learn.UnitaryLayer(alg),
                    learn.Residual(learn.ChannelLayer(alg, kraus_rank=2), p=0.5, learn_p=True),
                    learn.LindbladLayer(alg, n_jumps=1))
opt = torch.optim.Adam(net.parameters(), lr=0.05)
for step in range(200):
    loss = learn.infidelity(net(X), Y).mean()
    opt.zero_grad(); loss.backward(); opt.step()
```

Listing 4: A network of channels of $\mathcal{M} = M_1 \oplus M_3 \oplus M_2$, trained end to end with `torch.optim`. Every iterate is a channel.

backward of Section 6.5. Listing 4 shows a small network trained on input–output pairs; the losses provided are the infidelity, the trace distance, the relative entropy and the negative log-likelihood of measurement counts (from `POVM` of the module `measurement`).

Where the nonlinearity comes from. A composition of channels is again a channel, so depth adds no new maps: one Stinespring layer of sufficient width is universal, and a deep network of channels is a reparametrisation of that set, as a deep linear network is of the linear maps. As a function of the parameters the output is nonlinear, as a function of the input state it is linear; on encoded classical data this makes such a model a kernel method [147]. A genuinely nonlinear layer needs one of three physical resources: post-selection, whose renormalisation by the acceptance $\mathrm{Tr}(P\rho P)$ is the activation (`PostSelect` records the acceptance, whose logarithm is the natural penalty, since acceptances multiply along the network); several copies of the state; or a mean-field limit of many symmetric systems, whose finite-size dynamics the module `permutational` computes exactly in its total-spin sectors.

Skip connections. The residual block of a ResNet [141], $x \mapsto x + f(x)$, becomes $\mathrm{id} + \Phi$ for a channel Φ, which is completely positive but doubles the trace. Its normalised counterpart `Residual` is the convex combination $(1-p)\,\mathrm{id} + p\,\Phi$: one step of a jump process that applies Φ at rate p, so that n blocks with $p = t/n$ tend to $e^{t(\Phi - \mathrm{id})}$, the semigroup of the GKSL generator whose jump operators are the Kraus operators of Φ (verified in the test suite). ResNets become Lindblad evolutions, with the consequence studied in Section 13.2. Skips that copy activations forward (DenseNet, U-Net) have no counterpart for quantum states, which can neither be cloned nor broadcast [142]; only the commutative part can be copied. In the algebra this is the centre: an instrument writes its outcome into the centre of $\bigoplus_j \mathcal{M}$ (`measurement.Instrument.as_channel`), and `FeedForward` applies a branch per outcome, the quantum form of a fan-out connection. A block-diagonal channel $\bigoplus_k \Phi_k$ is likewise a mixture of experts whose router, the measurement of the sector, does not disturb the state.

Relation to other work. Feed-forward networks whose layers are completely positive maps were trained on quantum data by Beer et al. [137]; process tomography with learned models is done with tensor networks [144], and state tomography with neural-network states [143]; networks with parameters in a C^*-algebra were proposed in [138], and equivariant linear layers are elements of a commutant [48, 49]. What the library adds is the algebra: layers between algebras with different

sector structures, the centre as the carrier of classical information, losses defined through it, and the batching and gradients of the previous sections. The plan for this direction, with further experiments, is kept in the repository (`ROADMAP_learning.md`).

13.1 Tomography in the algebra as training

Process tomography by maximum likelihood is training in the sense of this section: the model is a parametrised channel, the data are measurement counts on the outputs of known inputs, and the loss is the negative log-likelihood. Learned models of this kind are used for process tomography with tensor networks [144] and for state tomography with neural-network states [143]. The algebra enters twice. First, it fixes what has to be measured. If $\mathcal{M} \subset B(\mathcal{H})$ is a unital $*$-subalgebra, then $\mathrm{Tr}(\rho A) = \mathrm{Tr}(E_{\mathcal{M}}(\rho)A)$ for $A \in \mathcal{M}$, where $E_{\mathcal{M}}$ is the trace-preserving conditional expectation, and the pairing $(\sigma, A) \mapsto \mathrm{Tr}(\sigma A)$ is nondegenerate on $\mathcal{M}$ (take $A = \sigma^*$). Measurements of observables of $\mathcal{M}$ therefore determine exactly $E_{\mathcal{M}}(\rho)$, and a process that maps $\mathcal{M}$ into $\mathcal{M}$, such as the noisy register of Section 12, is determined on $\mathcal{M}$ by inputs in $\mathcal{M}$ and measurements in $\mathcal{M}$. Permutationally invariant tomography [145] is the case of the commutant of the permutations. Second, the model can respect the sectors. A channel of $\mathcal{M} = \bigoplus_N M_{\binom{n}{N}}$ ranges over a set of dimension $(\dim \mathcal{M})^2 - \dim \mathcal{M}$, which is 4830 for $n = 4$, against $4^{2n} - 4^n = 65280$ for a channel of M_{2^n}. A GKSL generator whose jumps change N by at most one has fewer parameters still. We measure what each level of structure buys, with the same simulated experiment for all models.

Setup. The register is the ring of Section 12 with $n = 3$ and $n = 4$ qubits. The XY mixer acts for a time $T = 1$, together with amplitude damping, thermal excitation and dephasing at the rates $\gamma_\downarrow = 0.1$, $\gamma_\uparrow = 0.02$ and $\gamma_\varphi = 0.05$. The true process $\Phi = e^{T\mathcal{L}}$ is computed by `lindblad_evolve` with the exact exponential. A dense Lindblad evolution in M_{2^n}, with H and the jumps converted by `interop`, agrees with it to $1.9{\times}10^{-15}$. The training inputs are the 2^n basis states and $\binom{n}{N}^2$ Haar-random pure states in every sector with $\binom{n}{N} > 1$. They span $\mathcal{M}$, and there are 84 of them for $n = 4$. The held-out inputs are 17 further states: three pure states in each of these sectors and eight random mixed states with weight on all sectors. Each output is measured in 9 Haar-random bases of $\mathcal{M}$ (`POVM.from_unitary` of block-diagonal unitaries, combined with equal weights by `POVM.combine`). Every outcome of such a basis also reveals N, so the charge measurement `POVM.sectors` is a coarse-graining of each setting. The counts are sampled with 10^2 to 10^5 shots per input in total, and the exact probabilities serve as the limit of infinitely many shots. Four models are fitted:

(i) *rates*: the noise model with known jump operators and four unknowns $J, \gamma_\downarrow, \gamma_\uparrow, \gamma_\varphi$ (the coupling J multiplies the mixer);

(ii) *GKSL layer*: a `LindbladLayer` with a free self-adjoint $H \in \mathcal{M}$, $n - 1$ free jumps inside the sectors and n free jumps between neighbouring sectors $N \to N{\pm}1$; it contains the true generator;

(iii) *channel of $\mathcal{M}$*: a `ChannelLayer` from $\mathcal{M}$ to $\mathcal{M}$ with all sector transitions and Kraus rank $k^2_{\max}$, enough for every channel of $\mathcal{M}$;

(iv) *channel of M_{2^n}*: a `ChannelLayer` on M_{2^n} with Kraus rank 4^n, enough for every channel. This is the experimenter who ignores the symmetry: the same inputs, embedded by `interop.to_dense`, are measured in 19 Haar-random bases of M_{2^n}, with the same number of shots per input. The model has to learn that the process creates no coherences between sectors.

The raw parameter counts (`learn.count_parameters`, Table 23) include the Kraus gauge; the sets of channels that models (iii) and (iv) can reach have dimensions 4830 and 65280, and the inputs in

n	model	parameters	held-out trace distance, shots per input 10^2	10^3	10^4	10^5	exact	Choi error 10^4 shots	time [s]
3	rates	4	1.0×10^{-2}	2.8×10^{-3}	1.5×10^{-3}	4.8×10^{-4}	7.4×10^{-8}	1.7×10^{-3}	1
3	GKSL layer	280	9.0×10^{-2}	3.8×10^{-2}	1.6×10^{-2}	5.9×10^{-3}	4.2×10^{-6}	2.6×10^{-2}	8
3	channel of $\mathcal{M}$	1152	1.0×10^{-1}	4.0×10^{-2}	1.6×10^{-2}	7.2×10^{-3}	4.6×10^{-4}	3.0×10^{-2}	7
3	channel of M_{2^n}	8192	2.2×10^{-1}	9.0×10^{-2}	4.5×10^{-2}	1.9×10^{-2}	8.4×10^{-4}	9.2×10^{-2}	7
4	rates	4	7.4×10^{-3}	2.6×10^{-3}	9.6×10^{-4}	2.6×10^{-4}	3.6×10^{-9}	1.0×10^{-3}	4
4	GKSL layer	1386	1.2×10^{-1}	5.9×10^{-2}	2.2×10^{-2}	7.4×10^{-3}	1.3×10^{-5}	3.8×10^{-2}	51
4	channel of $\mathcal{M}$	18432	1.4×10^{-1}	6.3×10^{-2}	2.6×10^{-2}	1.0×10^{-2}	6.5×10^{-4}	5.2×10^{-2}	27
4	channel of M_{2^n}	131072	3.1×10^{-1}	1.4×10^{-1}	7.1×10^{-2}	3.3×10^{-2}	1.2×10^{-3}	1.7×10^{-1}	98

Table 23: Learning the noisy XY register from simulated counts. Parameters: real parameters of the model (`learn.count_parameters`, Kraus gauge included). Trace distance: mean over the held-out inputs of $\frac{1}{2}\|\hat{\Phi}(\rho) - \Phi(\rho)\|_1$, one data set per entry; "exact": fitted to the exact probabilities, which shows the optimisation error. Choi error: trace distance of the normalised Choi matrices of the restrictions to the input sectors, averaged over the sectors. Time: wall-clock seconds per random start, averaged over the shot budgets (one CPU core per fit).

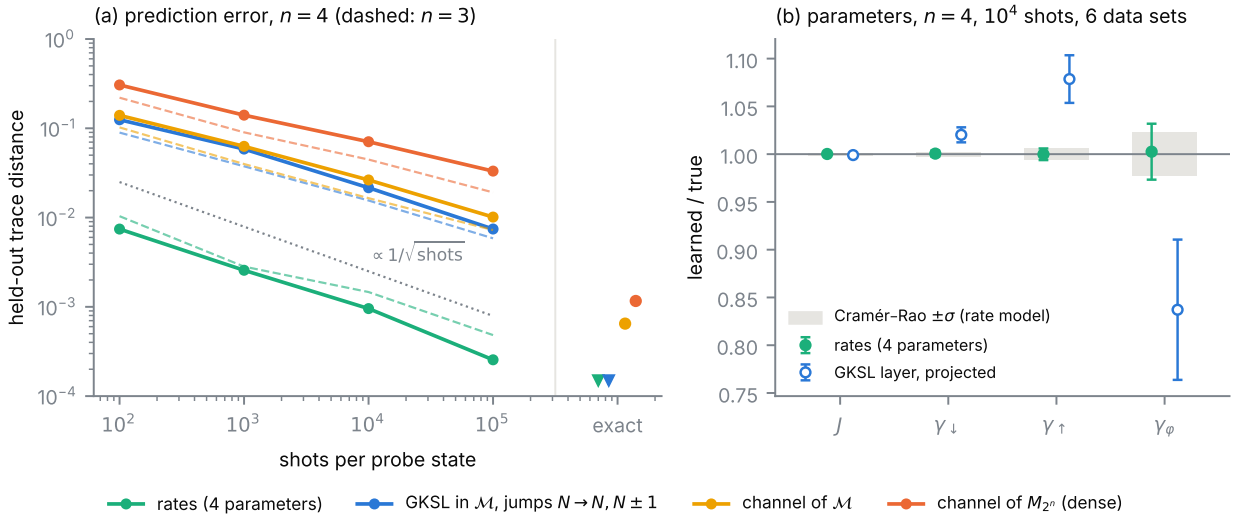


Figure 11: (a) Held-out trace distance against the number of shots per input for the four models, $n = 4$ (solid) and $n = 3$ (dashed); at the right, the fit to the exact probabilities (triangles: below 10^{-4}). (b) Learned coupling and rates for $n = 4$ at 1×10^4 shots per input: mean and standard deviation over 6 independent data sets, for the rate model and for the least-squares projection of the learned generator of the GKSL layer onto $-i[H, \cdot\,]$ and the three dissipators. Grey: the Cramér–Rao standard deviation of the rate model, from the Fisher information of the counts.

$\mathcal{M}$ identify 17850 of the latter. Each model is fitted by 300 Adam steps followed by at most 400 iterations of L-BFGS (both from `torch.optim`), with 3, 3, 2 and 2 random starts for models (i)–(iv), and the start with the smallest loss is kept. The losses of the starts agree to a relative 1.0×10^{-3} in every run with finite shots. During training the outputs are computed by equivalent fast paths: the exponential of the generator as a matrix on $\mathcal{M}$, of size $\dim\mathcal{M} = 70$, and Kraus operators applied to the pure input vectors. These agree with the forward passes of the layers to 5.7×10^{-16}. Evaluation uses the layers themselves. The whole study takes 14 minutes on 4 CPU cores.

Results. The models differ by large factors only where the structure is strong (Figure 11a). At 10^4 shots per input and $n = 4$, the held-out trace distance is 9.6×10^{-4} for the rate model, 2.2×10^{-2} for the GKSL layer, 2.6×10^{-2} for the channel of $\mathcal{M}$ and 7.1×10^{-2} for the dense channel. The rate model follows the law shots$^{-1/2}$ (fitted slope 0.48 over the four budgets). The larger models fall more slowly in this range, with slopes 0.41, 0.38 and 0.32, so they are not yet in their asymptotic regime and the gaps between them grow with the number of shots. The structure of the algebra is worth more than an order of magnitude in shots: the dense model needs 1.4×10^4 shots per input to reach the error 6.3×10^{-2} that the channel of $\mathcal{M}$ reaches with 1.0×10^3, a factor 14. A large part of its error has a simple origin. Its predictions contain coherences between sectors of trace norm 5.3×10^{-2} (at 10^4 shots), which the true process never creates and which the channel of $\mathcal{M}$ excludes by construction. The GKSL layer gains much less over the channel of $\mathcal{M}$ than its parameter count suggests: it needs 8.1×10^2 shots for the same error although it has 13 times fewer raw parameters. For so small a register, Markovianity and nearest-neighbour jumps constrain little. Knowing the jump operators is what pays: the rate model is below the target error already at 10^2 shots (7.4×10^{-3}). The fits to the exact probabilities reach 1.3×10^{-5}, 6.5×10^{-4} and 1.2×10^{-3} for models (ii)–(iv), far below the statistical errors, so the comparison at finite shots is not limited by the optimiser. The same ordering holds for the Choi error and for $n = 3$, where the GKSL layer and the channel of $\mathcal{M}$ tie at 10^4 shots.

Learned rates. For the rate model the learned values at 1×10^4 shots are $J = 1.0002 \pm 0.0015$, $\gamma_\downarrow = 0.1001 \pm 0.0003$, $\gamma_\uparrow = 0.0200 \pm 0.0001$ and $\gamma_\varphi = 0.0501 \pm 0.0015$ (mean and standard deviation over 6 data sets). The Cramér–Rao standard deviations, from the Fisher information of the counts at the true parameters (`measurement.fisher_information`), are 0.0013, 0.0003, 0.0001 and 0.0011, so the maximum-likelihood estimator is unbiased and close to efficient here (with six data sets the empirical spread is itself uncertain by about a third). The GKSL layer is not a rate estimator. Its generator, projected onto the physical terms, gives $\gamma_\downarrow = 0.1020 \pm 0.0008$, $\gamma_\uparrow = 0.0216 \pm 0.0005$ and $\gamma_\varphi = 0.0419 \pm 0.0037$. These are biased by more than their spread, and a fraction 3.1×10^{-2} of the learned generator lies outside the physical span: the free jumps absorb statistical noise as extra dissipation, which the projection attributes in part to the rates. This fraction falls with the number of shots (from about 0.19 at 10^2 to 0.01 at 10^5 shots per input, one data set each), and from the exact probabilities the GKSL layer recovers the rates to a relative 10^{-4} or better.

Scope. The data are simulated without state-preparation and measurement errors, the inputs and measurement bases are known exactly, and the true process lies inside every model class: it is Markovian and respects the sectors exactly. The experiment therefore measures statistical efficiency, not robustness to a wrong model. With noise that breaks the $U(1)$ symmetry, models (i)–(iii) would be biased and the dense model would not, which is where the dense model would win; we did not simulate this case. The dense comparison changes the measurement together with the model, as an experimenter without the symmetry would; we did not separate the two effects. The registers are small ($n \leq 4$, one evolution time), and every fit is a local optimisation. For the GKSL layer the start matters: with the library's default initial Hamiltonian of unit scale, only 1 of 3 starts reached the optimum for the exact data at $n = 4$, and the others stopped at losses up to 2.7×10^{-3}. We therefore start it near the identity channel, with the Hamiltonian scaled by 0.1, and with this choice all starts agree. The full-rank models cost one parameter per entry of a Stinespring isometry. At $n = 5$ the dense model would have $4^{10} \approx 10^6$ complex Kraus entries, which is why the study stops at $n = 4$; the GKSL layer and the channel of $\mathcal{M}$ would still be practical there.

13.2 Residual channel networks and Markovianity

A stack of residual blocks $(1-p_l)\,\mathrm{id}+p_l\Phi_l$ with small p_l is a product of channels close to the identity. Wolf and Cirac [146] call a channel *infinitesimal divisible* if it is a limit of channels that are, for every $\varepsilon>0$, products of channels ε-close to the identity; these include all products of Markovian channels $e^{\mathcal{L}}$, and their superoperators have nonnegative determinant. For residual blocks the determinant condition has a two-line proof, and it gives a quantitative obstruction.

The determinant obstruction. Write the superoperator T of a channel of M_d in a basis of self-adjoint matrices: a real $d^2\times d^2$ matrix with spectrum in the closed unit disc. If $p<\frac12$, the eigenvalues $1-p+p\lambda$ of $(1-p)\,\mathrm{id}+p\,T$ have modulus at least $1-2p$; the real ones are positive and the others come in conjugate pairs, so its determinant is at least $(1-2p)^{d^2}>0$. Determinants multiply and $\det e^{t\mathcal{L}}=e^{t\,\mathrm{tr}\,\mathcal{L}}>0$, so every residual network with $p<\frac12$ and every Lindblad layer has positive determinant, at any depth. The Pauli channel $\Phi_P(\rho)=\frac13(X\rho X+Y\rho Y+Z\rho Z)$ has Pauli transfer matrix $(R_{ab}=\frac12\,\mathrm{Tr}(\sigma_a\Phi(\sigma_b)),\ \sigma=1,X,Y,Z)$ equal to $\mathrm{diag}(1,-\frac13,-\frac13,-\frac13)$, with determinant $-\frac{1}{27}$. A trace-preserving map has $R=\left(\begin{smallmatrix}1&0\\ t&M\end{smallmatrix}\right)$, so $\det R=\det M$. If $\det M\ge 0$, the segment from $-\frac13\mathbf{1}$ to M contains a singular matrix, which by Eckart–Young is at distance at least $\frac13$ from $-\frac13\mathbf{1}$ in the Frobenius and in the operator norm. Since the output Bloch vectors differ by $t+(M+\frac13\mathbf{1})n$, every such map T satisfies

$$\|R_T-R_{\Phi_P}\|_F\ge\tfrac13,\qquad \sup_{\psi}\tfrac12\big\|T(\psi)-\Phi_P(\psi)\big\|_1\ge\tfrac16. \tag{4}$$

The second bound is sharp for infinitesimal divisible channels: a rotation by π about x after the Pauli semigroup with eigenvalues $(\varepsilon,\varepsilon,\frac13)$ is a product of two Markovian channels, and as $\varepsilon\to0$ it tends to $\mathrm{diag}(1,0,0,-\frac13)$, at worst-case trace distance $\frac16$ (and Frobenius distance 0.471). The best single semigroup we know, a rotation by π about z composed with a Pauli semigroup, $R=\mathrm{diag}(1,-a,-a,a^2)$, $a=0.2397$, is at Frobenius distance 0.413.

Experiment. In M_2 we fit three targets: (i) $e^{\mathcal{L}}$ for a random GKSL generator (a random Hamiltonian and two random jump operators); (ii) amplitude damping ($\gamma=0.5$) after a rotation by $2\pi/3$, a product of two Markovian channels that is not itself a semigroup. Its spectrum is non-degenerate, so by the test of Wolf, Eisert, Cubitt and Cirac [148] it is $e^{\mathcal{L}}$ only if a branch of its logarithm is conditionally completely positive. The principal branch misses by 0.253, and on the branch k the compressed Choi matrix is J_0+kG, so a vector with $\langle v,Gv\rangle<0$ (for $k>0$) or >0 (for $k<0$) excludes all branches with $|k|$ above an explicit bound; here the bound is 0, so no branch passes; and (iii) Φ_P. The models are (a) one `ChannelLayer` of Kraus rank four, which contains every qubit channel; (b) L blocks `Residual(ChannelLayer(alg, 4), p)` with $p=0.2$, $L\in\{1,2,4,8,16,32\}$ (a learned weight $p'\le p$ would add nothing, since the inner channel may itself be $(1-s)\,\mathrm{id}+s\Psi$); and (c) one `LindbladLayer` with a Hamiltonian and three jump operators, which contains every generator of M_2. The data are the outputs on $|0\rangle,|1\rangle,|+\rangle,|+i\rangle$, the loss is $\|R_{\mathrm{net}}-R\|_F^2$, and every model is trained by Adam for 2000 steps from 6 random starts. The starts run in parallel as the sectors of $\bigoplus_c M_2$: every layer is sector preserving with independent blocks, so each sector is an independent network. The study takes 281 s on 4 CPU cores (`scripts/residual_markov.py`).

Results. The Stinespring layer fits all three targets (errors 1.5×10^{-15}, 2.0×10^{-4}, 2.5×10^{-4}), Figure 12. On the Markovian targets the residual error decreases with depth, from 1.370 at $L=1$ to 2.5×10^{-15} at $L=32$ for (i), and from 1.561 to 0.163 for (ii). The Lindblad layer fits (i) to 1.4×10^{-4} but stops at 0.101 on (ii), which is not a semigroup. On the Pauli channel every fitted residual network

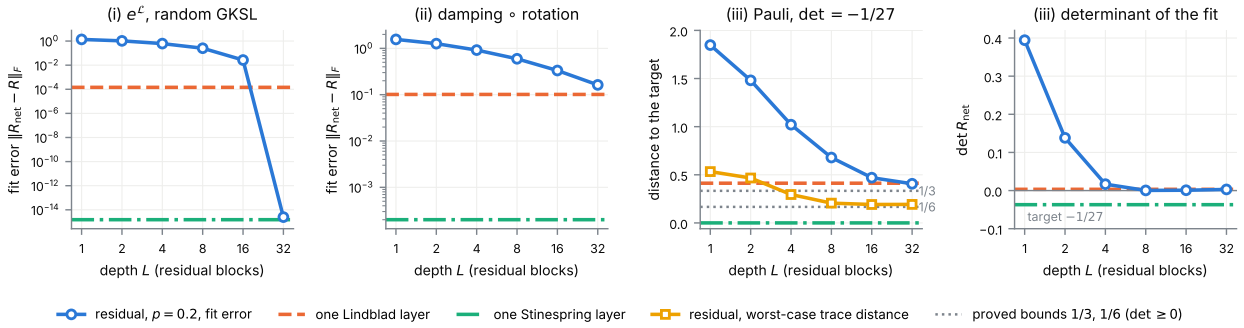


Figure 12: Fit error $\|R_{\text{net}} - R\|_F$ (best of 6 starts) against the depth L of a residual network with blocks $(1-p)\,\mathrm{id} + p\,\Phi_l$, $p = 0.2$ (circles), one Lindblad layer (dashed) and one Stinespring layer (dash-dotted), for (i) $e^{\mathcal{L}}$ with a random GKSL generator, (ii) amplitude damping after a rotation and (iii) the Pauli channel Φ_P, $\det = -\frac{1}{27}$. In (iii), squares: smallest worst-case trace distance among the fitted networks; dotted: the bounds $\frac{1}{3}$ and $\frac{1}{6}$ of (4). Right: determinant of the fitted maps for (iii).

and the Lindblad layer have positive determinant, the residual error never falls below 0.407, and the Lindblad layer stops at 0.413, the semigroup above. The smallest worst-case trace distance among the residual fits is 0.1916 (at $L = 16$): training minimises the Frobenius error, not this distance, whose infimum over infinitesimal divisible channels is $\frac{1}{6}$. At every depth the 6 starts of the residual network end within 4.0×10^{-5} of each other. Proved are (4) and the sign of the determinant; the floor of the Frobenius error, between $\frac{1}{3}$ and 0.407, is observed, not proved.

Reading. The three models are nested: semigroups, limits of residual networks with small blocks, and all channels. On the same data the two gaps between their errors measure two kinds of non-Markovianity: dependence on time, as for (ii), which a time-dependent master equation can still produce; and no Markovian master equation at all, not even a time-dependent one, of which $\det R < 0$ is an exact certificate, as for (iii). A residual fit error that stays away from zero as the depth grows is therefore a learned witness of non-Markovianity, comparable with the test of [148]. The witness is one-sided, since a large error can also be a failure of the optimiser; restarts and the Stinespring fit, which reaches the data, control this.

14 Applications in the companion papers

The library was built for, and used in, four further studies, which are reported with proofs, validation and figures in companion papers; here we only name the parts of the library they use.

How strongly trace characterisations fail (with S. A. Abed [1]). For 25 inequalities that characterise the trace, the violation by $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$ is maximised over all operators at every contrast $\Delta(W)$, in one batched optimisation per inequality (`optimize.SpectrumParam`, `PositiveParam`, `UnitaryParam`; lattice operations $P \wedge Q$, $P \vee Q$ by functional calculus with Daleckii–Krein gradients), and the best points are recomputed in mpmath with 40 digits. The computed envelopes led to closed forms, proved in [1], that vanish at the trace to the orders δ, δ^2, δ^4 and δ^8.

The Connes–Størmer entropy of two MASAs [2]. The relative entropy $H(\Delta_n | U\Delta_n U^*)$ of the diagonal algebra and a rotated copy is a supremum over partitions of unity; with the conditional

expectations of `Inclusion` and partitions parametrised by isometries, many unitaries and starts form one batch. For $n = 2$ the computation suggested a closed form, proved in [2]; for $n = 3, 4, 5$ it finds partitions that beat every orthonormal basis found.

A boundary time crystal under local dissipation [3]. Local emission and dephasing of N two-level systems commute with the permutations of the sites, so the state stays in the Schur–Weyl algebra $\bigoplus_j M_{2j+1} \otimes 1_{m_j}$, and the local dissipators are maps between its sectors (`PermutationalLindbladian`). With all sectors kept, the lifetimes of the oscillations follow for $N \leq 80$ without truncation, ten rates in one batch (a check-mode run on one CPU thread; the full mode, $N \leq 200$, is designed for a GPU and was not run).

Symmetry-resolved entanglement in the XXZ chain [3]. The reduced state of a charge-conserving state is an operator of $\bigoplus_q M_{d_q}$, so resolved entropies and charge statistics are read off its blocks. For $L = 24$ batched Lanczos runs, four anisotropies each, give the ground states for eight anisotropies, and the charge statistics give the Luttinger parameter to about one per cent.

15 Performance

Table 24 reports the wall-clock time per sample of library operations (not of isolated kernels), in single precision, with batch $B = \min\bigl(4096,\, 2^{22}/(Ck^2)\bigr)$ (full run on NVIDIA GeForce RTX 5090; CPU with 16 threads; PyTorch 2.14.1+cu130, CUDA 13.0, cuSOLVER 12004). The operations are: generating random operators with prescribed spectra, $f(A)$ by eigendecomposition, $|A|$, the Michelson contrast, applying a channel, the Lindblad channel $e^{t\mathcal{L}}$, and one gradient step of the optimisation.

Library versions matter: batched Hermitian eigensolvers on GPUs cover all block sizes only in recent cuSOLVER and PyTorch releases, and batched SVD on GPUs is limited to blocks of size at most 32, which is why the library uses the Hermitian eigensolver wherever it can. Batched QR on GPUs factorises small matrices one at a time, so Haar sampling on GPUs uses a batched Gram–Schmidt orthonormalisation with reorthogonalisation (the same matrices as QR with the phase correction). Table 24 was measured before this change; its rows "generate" on the GPU therefore show the earlier per-sector QR and understate the GPU. On the CPU the library splits batched decompositions into one chunk per thread, so all cores are used.

GPUs need enough independent work per kernel launch: many samples, many sectors or large blocks, and matrix products rather than decompositions of small matrices. The design follows from this. Everything is batched, so the batch dimension supplies the parallel work, and the cost module warns before configurations that would be slow. The structure of the computation is also checked directly (`scripts/gpu_check.py` on a GPU, `tests/test_sync_free.py` on a CPU): the operations read no values back to the host, and their number of kernel launches does not grow with C at fixed total size. For workloads that are small or dominated by a single eigenproblem, the CPU, or SciPy, is the right tool, and the library runs there unchanged.

16 Limitations and roadmap

- *Layout.* Blocks are padded to $k_{\max}$. For very different sector sizes, such as the binomial sectors of spin chains, one works in single sectors, and `SU2Chain` builds the block of one total spin without padding; a bucketed layout for operators on all sectors is not implemented yet. Matrix products of padded operators also multiply the padding: for the ten-qubit register of Section 12

operation	k	C	batch	GPU [μs]	CPU [μs]	speedup
generate	4	1	4096	63.5	1.90	0.0299
f(A) eigh	4	1	4096	0.160	3.06	19.2
|A| SVD	4	1	4096	0.388	5.27	13.6
contrast	4	1	4096	0.0979	1.44	14.7
channel	4	1	4096	0.116	0.186	1.60
exp(tL)	4	1	4096	2.05	34.1	16.6
opt. step	4	1	4096	2.76	7.78	2.82
generate	4	16	4096	995	28.1	0.0283
f(A) eigh	4	16	4096	0.514	44.7	86.9
|A| SVD	4	16	4096	4.17	80.7	19.4
contrast	4	16	4096	0.358	22.0	61.5
channel	4	16	4096	1.34	4.53	3.38
exp(tL)	4	16	1024	49.2	556	11.3
opt. step	4	16	4096	9.54	80.2	8.41
generate	4	256	1024	15710	534	0.0340
f(A) eigh	4	256	1024	9.01	742	82.3
|A| SVD	4	256	1024	66.3	1283	19.4
contrast	4	256	1024	5.63	359	63.7
channel	4	256	1024	20.9	65.0	3.12
exp(tL)	4	256	64	7743	15365	1.98
opt. step	4	256	1024	133	1604	12.0
generate	16	1	4096	60.9	17.3	0.283
f(A) eigh	16	1	4096	0.206	23.8	116
|A| SVD	16	1	4096	0.780	44.5	57.0
contrast	16	1	4096	0.137	10.9	79.5
channel	16	1	4096	0.129	1.72	13.4
exp(tL)	16	1	64	1149	13608	11.8
opt. step	16	1	4096	3.61	74.3	20.6
generate	16	16	1024	1032	290	0.281
f(A) eigh	16	16	1024	1.76	419	238
|A| SVD	16	16	1024	10.7	746	69.6
contrast	16	16	1024	1.26	182	145
channel	16	16	1024	1.63	65.0	39.8
exp(tL)	16	16	4	79221	181417	2.29
opt. step	16	16	1024	34.9	1505	43.1
generate	16	256	64	18838	5937	0.315
f(A) eigh	16	256	64	78.0	6736	86.3
|A| SVD	16	256	64	172	11860	68.9
contrast	16	256	64	36.9	2871	77.7
channel	16	256	64	26.3	989	37.6
exp(tL)	16	256	1	2349043	3558629	1.51
opt. step	16	256	64	567	21657	38.2
generate	64	1	1024	62.0	329	5.30
f(A) eigh	64	1	1024	3.47	393	113
|A| SVD	64	1	1024	13.1	820	62.4
contrast	64	1	1024	2.48	211	85.0
channel	64	1	1024	0.859	71.2	82.9
opt. step	64	1	1024	34.2	1737	50.8
generate	64	16	64	1262	4995	3.96
f(A) eigh	64	16	64	59.2	6495	110
|A| SVD	64	16	64	211	12966	61.4
contrast	64	16	64	41.2	3424	83.0
channel	64	16	64	13.7	1054	76.8
opt. step	64	16	64	550	28154	51.2
generate	64	256	4	132631	103301	0.779
f(A) eigh	64	256	4	1753	99206	56.6
|A| SVD	64	256	4	3398	204740	60.2
contrast	64	256	4	933	54526	58.4
channel	64	256	4	217	12346	57.0
opt. step	64	256	4	8867	448973	50.6

Table 24: Wall-clock time per sample of library operations for C sectors of size k and the batch size shown (full run on NVIDIA GeForce RTX 5090, CPU with 16 threads) (single precision; "exp(tL)" uses a smaller batch because its superoperators have k^4 entries). With a GPU the table also gives the CPU time and the speedup CPU/GPU. The run predates the batched Gram–Schmidt generator of Haar unitaries on GPUs.

the dense Hamiltonian part of a time step costs about four times $\sum_N k_N^3$, and it, not the jumps, dominates the run time.

- *Symmetries.* Abelian charges with an arbitrary fusion rule, lattice translations and, for spin-$\frac{1}{2}$

chains, SU(2) multiplets are supported; SU(2) with momentum, larger spins, reflections, spin flip and sparse solvers in SU(2) blocks are not. The Wedderburn decomposition uses dense linear algebra. Dense methods reach about 16–18 spins in the $U(1)$ sectors, the sparse solvers the half-filled sector of 24 spins (dimension $2.7 \cdot 10^6$).

- *Noise.* Only noise that commutes with the symmetry, in the weak sense of Section 12, can be represented; bit flips and other symmetry-breaking errors leave the algebra.
- *Optimisation.* The search is local, so envelopes are lower bounds for suprema. Gradients of the functional calculus are finite at degenerate spectra, but gradients through the eigenvectors returned by `eigh` are not, and second derivatives of $f(A)$ are correct at non-degenerate spectra only.
- *Divergences and scale.* For small α the sandwiched divergence of ill-conditioned states is not accurate (hence $\alpha \geq 0.2$ in Section 9.1). Choi matrices, superoperators and $e^{t\mathcal{L}}$ work with $k_c^2 \times k_c^2$ matrices and are limited to blocks of a few tens; RK4 and the splitting scale further.
- *Learning.* The layers of Section 13 are small: dense Kraus blocks and dense generators, trained on simulated data for registers of a few qubits. Encoders into smaller algebras, tomography of larger registers with sparse measurements, training viewed as a dynamics of the channel and certified upper bounds for the learned maps are not implemented; the plan is kept in the repository (`ROADMAP_learning.md`).
- *Scope.* Type I with finite-dimensional fibres: a continuous centre only through the discretisation of Section 11.1, infinite factors only through truncation, and types II and III only through the finite stages of towers of inclusions; no tensor-network methods. Sparse methods exist only for sector Hamiltonians and for jumps that map basis vectors to basis vectors; other jumps between sectors are stored densely. Results are cross-checked against QuTiP and QuSpin in the test suite, and the permutational Lindblad solver is timed against QuTiP's PIQS in [3]; systematic benchmarks against QuTiP, `dynamiqs`, QuantumToolbox.jl, QuSpin and XDiag are future work.

17 Availability and reproducibility

The code is available under the MIT licence at `https://github.com/andre-a-no/von_neumann_type_i`; this paper describes version 1.2.0 (tag `v1.2.0`) with the numerical corrections listed under `Unreleased` in `CHANGELOG.md`, and its scripts need the repository version of the library (`pip install -e ".[scripts]"`) rather than an older release from PyPI. The command `bash scripts/run_all.sh full cuda` regenerates every table and figure on a GPU; without the mode argument it runs a reduced version on the default device, which takes well under an hour on a four-core CPU. The script `scripts/gpu_check.py` checks a GPU, and `scripts/library_paper_schematics.py` draws the schematic figures and the summary figure of the noisy register. The tables state the device and, except the use-case tables of Sections 12 and 12.1, which are single full-size runs, the mode they were produced with; the JSON files written by the current versions of the scripts also record the commit of the code. The stored tables and figures retain the original runs, which predate these corrections; they have not been recomputed for the corrected implementation. Some provenance records contain `+dirty`, which does not identify the uncommitted source changes. They therefore do not establish exact reproducibility from the recorded commit alone. `man/REPRODUCIBILITY.md` records this status and the affected pipelines; a new numerical release requires runs from an identified clean revision and an explicit comparison with the stored results.

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